POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Study of Depolarization Effects on Polarization Entanglement Distribution through Optical Fibers

Tesi di Laurea Magistrale in
High Performance Computing Engineering

Author: **Marco Cioci**

Student ID: 252488
Advisor: Prof. Paolo Martelli
Co-advisors: Dr. Alessandro Gagliano
Academic Year: 2025–2026

Abstract

As quantum computing technologies continue to advance and quantum communication links become increasingly relevant for interconnecting remote quantum systems, the reliable distribution of quantum states over optical fibers is becoming a key technological requirement.

Polarization-entangled photon pairs are well suited to fiber links because polarization can be manipulated and measured with standard optical components. Real fibers, however, introduce coherent transformations and decoherence mechanisms that modify the observed correlations and may degrade the distributed entanglement. This thesis develops a progressive framework connecting these physical effects to numerical predictions and experimentally accessible quantities.

The modeling starts from an experimentally motivated description of the residual polarization evolution remaining after partial compensation. A fixed relative phase is first introduced to distinguish coherent changes in measured correlations from actual entanglement degradation. The model is then extended to unresolved phase fluctuations, summarized by the coherence parameter $\mu = \langle e^{i\theta} \rangle$, whose magnitude determines the residual concurrence within this model through $C = |\mu|$. Independent depolarizing effects in the two fiber arms are subsequently introduced through the correlation-survival factor $\eta = (1-p_A)(1-p_B)$ and combined with phase fluctuations in a common density-operator description. Finally, the effective models are connected to polarization-mode dispersion, where finite-bandwidth photons undergo frequency-dependent birefringent transformations and polarization mixedness emerges after tracing over unresolved spectral degrees of freedom.

The framework is implemented in a standalone modular MATLAB library reproducing this modeling hierarchy numerically. The simulator provides a common pipeline for state preparation, channel composition, statistical averaging, polarization-correlation analysis, and evaluation of purity, fidelity, concurrence, and Bell-inequality parameters. Analytical predictions are compared with numerical results to validate the implementation and identify the different signatures of coherent evolution, statistical dephasing, depolarization, and PMD.

The theoretical and computational work is complemented by the development of a two-arm laboratory platform for polarization-entanglement distribution based on single-photon detection and coincidence measurements together with controllable polarization transformations. The setup provides the experimental counterpart of the developed framework and a basis for a future entanglement-based BBM92 quantum-key-distribution implementation. Overall, the thesis establishes a continuous path from fiber-channel modeling, through numerical simulation, to the experimental characterization of entanglement distribution in optical fibers.

Keywords: Polarization Entanglement, Optical Fibers, Quantum Channels, Decoherence, Depolarization.

Abstract in lingua italiana

Parole chiave: Entanglement di Polarizzazione, Fibre Ottiche, Canali Quantistici, Decoerenza, Depolarizzazione.

Contents

Introduction

As quantum computing technologies continue to advance and quantum communication links become increasingly relevant for interconnecting remote quantum systems, the reliable transmission of quantum states between distant nodes is becoming an increasingly important technological requirement. Optical fibers represent the natural transmission medium for many such applications because of their compatibility with existing telecommunications infrastructure and their suitability for long-distance photonic links [1].

Among the different photonic degrees of freedom, polarization provides a particularly convenient encoding for quantum information. Polarization-entangled photon pairs can be generated, manipulated, and measured using established optical components, making them attractive resources for entanglement distribution and entanglement-based quantum communication protocols [2, 3]. Propagation through real optical fibers, however, modifies the polarization state through birefringence, polarization rotations, phase accumulation, temporal effects, and environmental fluctuations [4].

Some of these transformations remain coherent and therefore preserve the entanglement of the state, whereas unresolved fluctuations or coupling to additional degrees of freedom can lead to effective decoherence and degradation of the observed polarization correlations [5, 6].

In this work, the term *polarization decoherence* is used in a general sense to describe the effective loss of coherence of the polarization degree of freedom when information becomes inaccessible to the polarization measurement. This may occur either because different experimental realizations undergo unresolved fluctuating transformations or because polarization becomes coupled to additional degrees of freedom, such as frequency or arrival time, which are subsequently traced out. Although the complete physical evolution may remain unitary, the resulting reduced polarization state can therefore become mixed.

Two more specific forms of polarization degradation are distinguished within this general framework. *Dephasing* denotes the suppression of relative phase coherence between selected polarization components while leaving their populations unchanged. In the density-operator representation, this appears as a reduction of the corresponding off-diagonal matrix elements. Random fluctuations of a relative phase provide the simplest example considered in this thesis. By contrast, *depolarization* denotes a reduction of the degree of polarization without selecting a particular phase relation. For a single qubit, this corresponds geometrically to a contraction of the Bloch vector toward the center of the Bloch ball. In the effective channel models developed here, depolarization is represented through isotropic local depolarizing channels acting independently on the two fiber arms. Dephasing and depolarization are therefore treated as distinct effective processes, although both constitute forms of reduced polarization decoherence and both may lead to degradation of bipartite correlations and entanglement.

It is also important to distinguish these irreversible or effectively irreversible processes from deterministic birefringent evolution. A fixed polarization rotation or a fixed differential phase modifies the orientation and correlation structure of the state but, as a local unitary transformation, does not by itself reduce its purity or entanglement. Loss of polarization coherence appears only when the underlying transformation is not resolved at the level of the measurement or when the polarization degree of freedom becomes correlated with unobserved variables. This distinction between coherent evolution and effective decoherence constitutes one of the central organizing principles of the modeling strategy adopted in this work.

The aim of this thesis is to study these effects through a framework connecting physical modeling, numerical simulation, and laboratory characterization of polarization-entangled photon pairs propagating through two independent fiber links. Rather than introducing a complete fiber model from the beginning, the problem is approached through a progressive modeling strategy. Each physical mechanism is first isolated in an analytically tractable description and subsequently combined with additional effects, allowing its influence on the propagated quantum state to be identified independently and the corresponding numerical implementation to be validated.

The physical reference system consists of a source producing polarization-entangled photon pairs, whose photons propagate along two spatially separated arms, denoted A and B. The polarization degree of freedom is used as the qubit encoding, with $|H\rangle \equiv |0\rangle$ and $|V\rangle \equiv |1\rangle$. The ideal input state considered throughout most of the work is the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + |10\rangle \right).$$

Fiber propagation acts locally on the two photons, while the resulting state is characterized through its reduced states, polarization correlations, purity, fidelity, concurrence, and Bell-nonlocality indicators.

The first level of the propagation model is motivated directly by the polarization-compensation procedure available in the experimental setup. After the dominant polarization rotation has been compensated, a residual relative phase between the two polarization components may remain. This effect is represented through the reduced phase model and provides the simplest description for studying how coherent fiber evolution modifies the measured two-photon correlations without degrading the entanglement itself.

The description is then extended to unresolved phase fluctuations by averaging over multiple phase realizations. In this regime, the complex parameter

$$\mu = \langle e^{i\theta} \rangle$$

summarizes the surviving coherence and provides a direct connection between the statistics of the phase fluctuations and the properties of the resulting mixed state. This model constitutes a controlled example of polarization dephasing: the populations in the selected basis remain unchanged, while the corresponding off-diagonal coherence terms are progressively suppressed.

A second modeling level introduces depolarizing effects in the two propagation arms. Since the photons travel through physically distinct fibers, these effects are treated through independent local channels acting on subsystems A and B. Unlike the reduced-phase and dephasing models, which act primarily on relative phase coherence, the depolarizing channel contracts the complete local polarization information isotropically. Its combined action with the residual phase evolution is then studied through an effective two-arm channel in which coherent evolution, statistical dephasing, and local depolarization can be distinguished within the same density-operator framework.

The effective description is finally connected to a more physical treatment of polarization-mode dispersion (PMD) [7–9]. In this case, the finite spectral bandwidth of the photons is taken into account explicitly and different frequency components experience different birefringent transformations. The resulting evolution can correlate polarization with frequency and arrival time. Although the complete polarization–frequency state remains pure under lossless unitary propagation, tracing over unresolved spectral degrees of freedom can produce an effectively mixed polarization state. PMD therefore provides a physical mechanism by which polarization decoherence can emerge from an underlying unitary evolution. A segmented birefringent-fiber representation is introduced to model this frequency-dependent propagation and to connect the reduced channel descriptions with the physical origin of polarization decoherence in optical fibers.

These models are implemented in a standalone modular MATLAB library. The computational framework follows the same hierarchy as the theoretical development: state preparation, operator construction, propagation, statistical averaging, quantum-channel application, measurement evaluation, and computation of state metrics are implemented as separate reusable components. Numerical experiments are then constructed by combining these validated modules. Analytical predictions are used whenever available as reference solutions, so that increasingly complex models are introduced only after their constituent components have been independently verified.

The theoretical and computational activity is complemented by the development of a two-arm experimental platform for polarization-entanglement distribution. The setup combines controllable polarization transformations with single-photon detection and coincidence measurements, allowing the correlations predicted by the models to be related to directly accessible laboratory quantities. The experimental activity is intended both as a first comparison between the developed framework and a physical fiber link and as a basis for subsequent applications to entanglement-based quantum communication, including a future BBM92 implementation.

The thesis is organized as follows. Chapter 1 introduces the physical and mathematical framework used throughout the work, including the bipartite polarization system, density-operator formalism, reduced states, correlation tensors, local measurements, and the metrics used to quantify state quality, entanglement, and nonlocality. Chapter 2 develops the fiber-propagation models, progressing from reduced phase evolution and statistical phase fluctuations to local depolarizing channels, composite effective channels, and the frequency-dependent description of polarization-mode dispersion. Chapter **??** presents the MATLAB simulation framework and the numerical experiments used to validate the analytical models and investigate their predictions. Chapter **??** describes the experimental platform, measurement procedure, and laboratory activity developed for the distribution and characterization of polarization-entangled photon pairs. The final chapter **??** summarizes the main results and discusses possible extensions of the modeling, simulation, and experimental framework.

1 | Mathematical and Physical Framework

This chapter introduces the physical representation and mathematical tools used throughout the thesis to describe polarization-entangled photon pairs and their evolution through optical-fiber channels. The treatment is formulated within the density-operator framework of quantum information theory, which provides a common description for pure states, statistical mixtures, local measurements, and noisy quantum evolution [10].

The chapter first defines the bipartite polarization system and the reference Bell state. The density-operator and reduced-state formalisms are then introduced, followed by the representation of two-photon correlations through Pauli operators and the correlation tensor. Finally, the quantities used throughout the numerical and experimental analysis are defined, including purity, fidelity, concurrence, tangle, entanglement of formation, and CHSH nonlocality.

1.1. Bipartite Polarization System

The physical system considered in this work consists of a pair of polarization-entangled photons propagating along two spatially separated optical paths, denoted by A and B. In the experimental configuration considered later in this thesis, the photon pairs are produced through spontaneous parametric down-conversion (SPDC).

The quantum information carried by each photon is encoded in its polarization degree of freedom. Horizontal and vertical polarization are identified with the computational basis according to

$$|H\rangle \equiv |0\rangle, \qquad |V\rangle \equiv |1\rangle. \tag{1.1}$$

Each photon therefore defines a two-dimensional Hilbert space,

$$\mathcal{H}_A \simeq \mathbb{C}^2, \qquad \mathcal{H}_B \simeq \mathbb{C}^2. \tag{1.2}$$

The joint system is described in the tensor-product Hilbert space

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B \simeq \mathbb{C}^2 \otimes \mathbb{C}^2, \tag{1.3}$$

which has dimension four.

The ordered computational basis adopted throughout the thesis is

$$\mathcal{B}_{AB} = \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}, \tag{1.4}$$

where the first entry refers to subsystem A and the second to subsystem B. In polarization notation, the same basis is

$$\{|H\rangle_A|H\rangle_B, |H\rangle_A|V\rangle_B, |V\rangle_A|H\rangle_B, |V\rangle_A|V\rangle_B\}.$$

Consequently, pure states of the complete system are represented by four-component complex vectors and operators acting on both photons by 4×4 matrices. This basis convention is used consistently in the analytical derivations and in the MATLAB implementation.

1.2. Reference Entangled State

The principal input state considered in this work is the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right), \tag{1.5}$$

or equivalently,

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|H\rangle_A|V\rangle_B + |V\rangle_A|H\rangle_B\right). \tag{1.6}$$

The correspondence between the polarization basis and the standard single-qubit Bloch-sphere representation is illustrated in Fig. 1.1. In particular, $|H\rangle \equiv |0\rangle$ and $|V\rangle \equiv |1\rangle$, while coherent superpositions of the two basis states lie on the equator of the sphere.

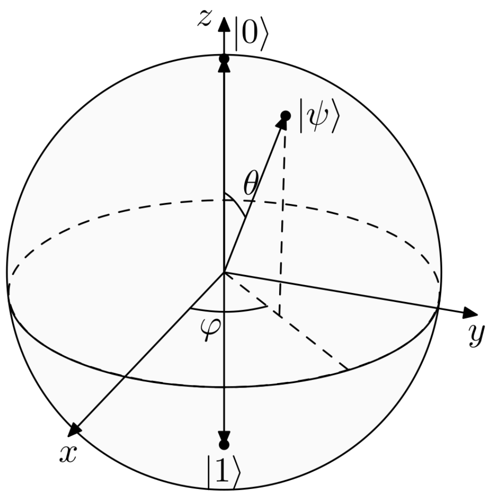

Figure 1.1: Bloch-sphere representation of a single polarization qubit. The horizontal and vertical polarization states correspond to $|H\rangle \equiv |0\rangle$ and $|V\rangle \equiv |1\rangle$.

The correspondence between the polarization basis and the standard single-qubit Bloch-sphere representation is illustrated in Fig. 1.1. In particular, $|H\rangle \equiv |0\rangle$ and $|V\rangle \equiv |1\rangle$, while coherent superpositions of the two basis states lie on the equator of the sphere.

In the ordered basis of Eq. (1.4), its vector representation is

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}. \tag{1.7}$$

The state $|\Psi^+\rangle$ is maximally entangled and therefore provides both a physically relevant resource and a convenient analytical benchmark for studying fiber-induced transformations.

For completeness, the four Bell states are:

$$|\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}, \qquad |\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}. \tag{1.8}$$

Although the theoretical framework applies to arbitrary two-qubit states, $|\Psi^+\rangle$ is used as the reference state throughout most of the simulations and laboratory analysis.

1.3. Density-Operator Description

A pure quantum state can be represented either by its state vector $|\psi\rangle$ or by the corresponding density operator

$$\rho = |\psi\rangle\langle\psi|. \tag{1.9}$$

For a bipartite system, the complete state is denoted by ρ_{AB}. The density-operator formulation is equivalent to the state-vector description for pure states, but it extends naturally to situations in which the system is mixed.

A general mixed state may be expressed as

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|, \qquad p_k \geq 0, \qquad \sum_k p_k = 1. \tag{1.10}$$

Such states arise naturally when different physical realizations are averaged statistically or when degrees of freedom correlated with the system are not resolved experimentally. Importantly, a density operator does not in general possess a unique ensemble decomposition; different statistical ensembles may represent the same physical state [10].

The density-operator formalism is therefore adopted as the common representation throughout the simulator.

1.3.1. Expectation Values

For a pure state, the expectation value of an observable O is

$$\langle O\rangle = \langle\psi|O|\psi\rangle.$$

In density-operator notation, the corresponding general expression is

$$\langle O\rangle = \mathrm{Tr}(\rho O). \tag{1.11}$$

For $\rho = |\psi\rangle\langle\psi|$, Eq. (1.11) reduces directly to the pure-state expression. The same equation remains valid for mixed states and therefore provides the basis for the evaluation of all observables used in this work.

1.3.2. Reduced Density Operators

For a bipartite state ρ_{AB}, the state accessible locally to either subsystem is obtained through the partial trace,

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}), \qquad \rho_B = \mathrm{Tr}_A(\rho_{AB}). \tag{1.12}$$

For example, tracing over subsystem B gives

$$\rho_A = \sum_{j=0}^{1} (I_A \otimes \langle j|)\, \rho_{AB}\, (I_A \otimes |j\rangle)\, . \tag{1.13}$$

The reduced state contains all information required to predict measurements performed exclusively on the corresponding subsystem. For an observable O_A,

$$\mathrm{Tr}\left[\rho_{AB}\,(O_A \otimes I_B)\right] = \mathrm{Tr}(\rho_A O_A). \tag{1.14}$$

For the Bell state of Eq. (1.5), the two reduced density operators are

$$\rho_A = \rho_B = \frac{I_2}{2}. \tag{1.15}$$

Thus, each photon is locally maximally mixed even though the complete two-photon state is pure and maximally entangled. The information distinguishing the Bell state from an uncorrelated local mixture is therefore contained in the joint correlations rather than in the individual reduced states.

1.3.3. Bloch Representation

Any single-qubit density operator can be written as

$$\rho = \frac{1}{2}\left(I + \mathbf{r} \cdot \boldsymbol{\sigma}\right), \tag{1.16}$$

where

$$\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$$

is the vector of Pauli matrices and $\mathbf{r} \in \mathbb{R}^3$ is the Bloch vector.

Physical states satisfy

$$\|\mathbf{r}\| \leq 1. \tag{1.17}$$

Pure states satisfy $\|\mathbf{r}\| = 1$ and lie on the surface of the Bloch sphere, whereas mixed states satisfy $\|\mathbf{r}\| < 1$. The maximally mixed state corresponds to

$$\mathbf{r} = 0, \qquad \rho = \frac{I_2}{2}.$$

Unitary transformations rotate the Bloch vector while preserving its norm. Effective decohering processes may instead reduce its magnitude, corresponding to motion toward the interior of the Bloch ball.

1.4. Quantum Evolution and Quantum Channels

The evolution of the two polarization qubits can occur either coherently or through an effective noisy process. These two regimes are naturally distinguished within the density-operator formalism.

1.4.1. Local Unitary Evolution

When each photon undergoes a coherent polarization transformation, its evolution is represented by a unitary operator acting on the corresponding subsystem.

If U_A and U_B denote the transformations acting on arms A and B, respectively, the pure state evolves according to

$$|\psi_{\text{out}}\rangle = (U_A \otimes U_B)|\psi_{\text{in}}\rangle. \tag{1.18}$$

At the density-operator level,

$$\rho_{\text{out}} = (U_A \otimes U_B)\rho_{\text{in}}(U_A^\dagger \otimes U_B^\dagger). \tag{1.19}$$

Local unitary evolution preserves global purity and bipartite entanglement. It may nevertheless modify the correlations observed with respect to fixed laboratory measurement

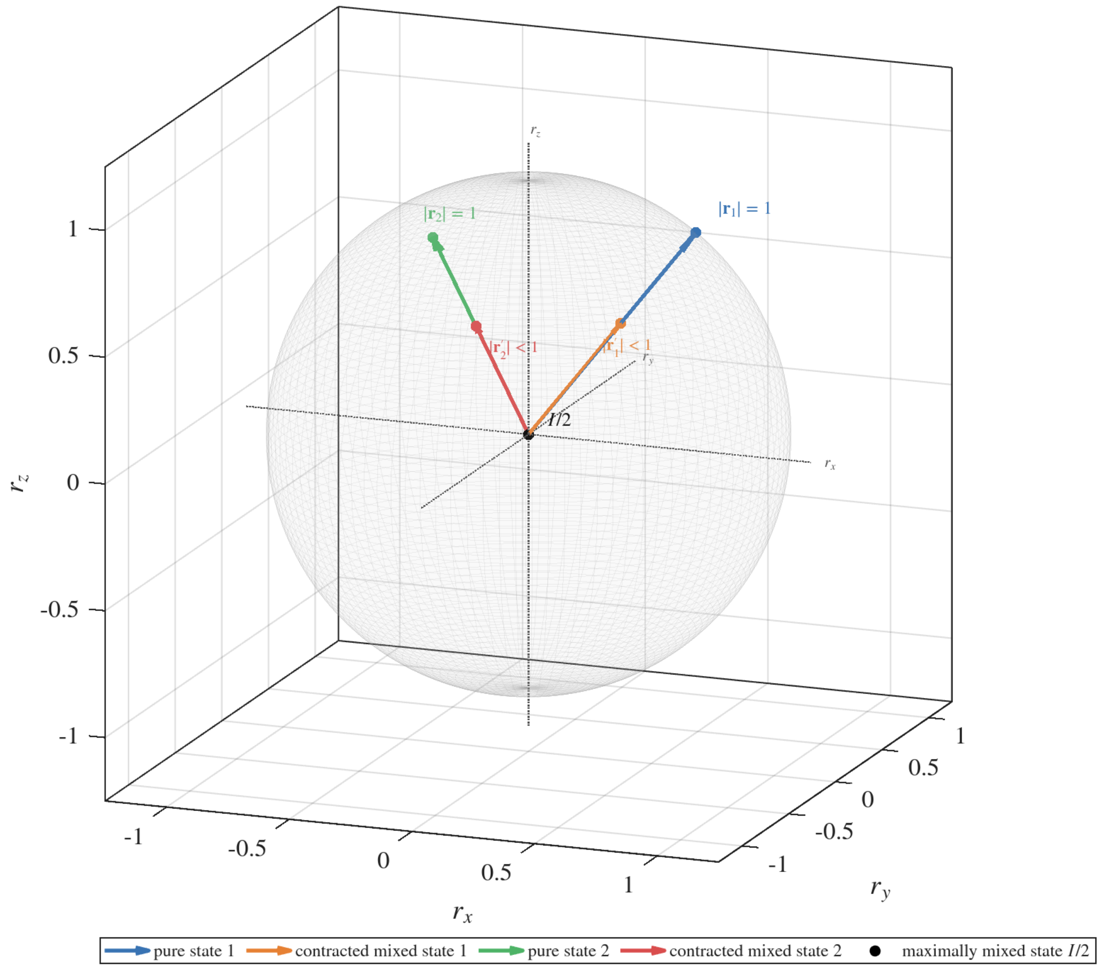

Figure 1.2: Bloch-sphere representation of pure and mixed single-qubit states. Pure states satisfy $\|\mathbf{r}\| = 1$, whereas mixed states are represented by vectors inside the Bloch sphere. The maximally mixed state corresponds to the origin.

axes. This distinction between a coherent transformation and genuine entanglement degradation is central to the fiber models developed in Chapter 2.

1.4.2. Quantum Channels

A more general transformation of a quantum state is represented by a quantum channel,

$$\mathcal{E} : \rho \mapsto \mathcal{E}(\rho), \tag{1.20}$$

which is a completely positive and trace-preserving linear map.

A general channel may be written in operator-sum form as [10, 11]

$$\mathcal{E}(\rho) = \sum_\mu K_\mu \rho K_\mu^\dagger, \qquad \sum_\mu K_\mu^\dagger K_\mu = I, \tag{1.21}$$

where the operators K_μ are the Kraus operators associated with the map.

Unitary evolution is recovered as the special case containing a single Kraus operator $K = U$. More general channels allow the reduced state to become mixed and are therefore required to describe decoherence, depolarization, and other effective noise mechanisms.

The specific channel models used to represent optical-fiber propagation are introduced in Chapter 2.

1.5. Polarization Correlations and Measurements

The information carried by a polarization-entangled state is primarily contained in joint measurement statistics. A convenient representation of these correlations is obtained using tensor products of Pauli operators.

1.5.1. Correlation Tensor

For a two-qubit state ρ_{AB}, the correlation tensor is defined by

$$T_{ij} = \mathrm{Tr}\left[\rho_{AB}\left(\sigma_i \otimes \sigma_j\right)\right], \qquad i, j \in \{x, y, z\}. \tag{1.22}$$

The tensor $T \in \mathbb{R}^{3\times3}$ therefore collects the nine bipartite Pauli correlations.

Introducing $\sigma_0 = I$, a general two-qubit state can be expanded as

$$\rho_{AB} = \frac{1}{4} \sum_{\mu,\nu \in \{0,x,y,z\}} S_{\mu\nu} \sigma_\mu \otimes \sigma_\nu, \tag{1.23}$$

where

$$S_{\mu\nu} = \mathrm{Tr}\left[\rho_{AB}\left(\sigma_\mu \otimes \sigma_\nu\right)\right]. \tag{1.24}$$

The quantities $S_{\mu\nu}$ correspond to generalized Stokes parameters. The terms S_{i0} and S_{0j} describe the local Bloch vectors, whereas the 3×3 block S_{ij} coincides with the correlation tensor T.

This representation is particularly useful in polarization experiments because the density operator can be reconstructed from expectation values of local Pauli measurements, as in standard two-qubit quantum-state tomography [12].

The reconstruction follows from the orthogonality relation

$$\mathrm{Tr}\left[(\sigma_i \otimes \sigma_j)(\sigma_k \otimes \sigma_l)\right] = 4\,\delta_{ik}\delta_{jl}, \qquad i,j,k,l \in \{0,x,y,z\}. \tag{1.25}$$

1.5.2. Arbitrary Local Measurement Axes

A projective qubit measurement along a unit vector $\mathbf{n} \in \mathbb{R}^3$ is represented by the observable

$$\sigma(\mathbf{n}) = \mathbf{n} \cdot \boldsymbol{\sigma} = n_x\sigma_x + n_y\sigma_y + n_z\sigma_z, \qquad \|\mathbf{n}\| = 1. \tag{1.26}$$

The normalization condition ensures that the observable has eigenvalues ± 1.

For measurement directions $\mathbf{a}$ and $\mathbf{b}$ on subsystems A and B, respectively, the two-photon correlation is

$$E(\mathbf{a},\mathbf{b}) = \mathrm{Tr}\left[\rho_{AB}\left(\sigma(\mathbf{a}) \otimes \sigma(\mathbf{b})\right)\right]. \tag{1.27}$$

Using Eq. (1.22), this becomes

$$E(\mathbf{a},\mathbf{b}) = \mathbf{a}^{\mathsf{T}} T \mathbf{b}. \tag{1.28}$$

The correlation tensor therefore determines the expectation value associated with any pair of local Pauli measurement directions.

Experimentally, measurements along arbitrary axes can be realized by applying appropriate local polarization transformations before a fixed polarization analysis stage. This provides the connection between the abstract operator description and the wave-plate and polarization-controller settings used in the laboratory.

1.6. Quantitative Characterization of the State

No single scalar quantity completely characterizes a bipartite quantum state. Different measures are therefore used in this work to distinguish mixedness, proximity to a target state, entanglement, and Bell nonlocality.

1.6.1. Purity

The purity of a density operator is defined as:

$$\gamma(\rho) = \mathrm{Tr}(\rho^2). \tag{1.29}$$

If ρ acts on a d-dimensional Hilbert space,

$$\frac{1}{d} \leq \gamma(\rho) \leq 1. \tag{1.30}$$

The upper bound corresponds to a pure state, whereas the lower bound is attained by the maximally mixed state I_d/d.

If

$$\rho = \sum_i \lambda_i |i\rangle\langle i|$$

is the spectral decomposition of the state, then

$$\gamma(\rho) = \sum_i \lambda_i^2. \tag{1.31}$$

For a single qubit written in Bloch form, Eq. (1.16) gives

$$\gamma(\rho) = \frac{1 + \|\mathbf{r}\|^2}{2}. \tag{1.32}$$

Purity quantifies mixedness but does not identify its physical origin. A mixed reduced state may result, for example, from tracing over an entangled subsystem, whereas a mixed global state may arise from statistical averaging or interaction with unresolved degrees of freedom.

1.6.2. Fidelity

Throughout this work, fidelity is defined using the squared Uhlmann convention, for two states ρ and σ it is given by

$$F(\rho,\sigma) = \left[\mathrm{Tr}\left(\sqrt{\sqrt{\rho}\,\sigma\,\sqrt{\rho}}\right)\right]^2. \tag{1.33}$$

When the reference state is pure, $\sigma = |\phi\rangle\langle\phi|$, the expression simplifies to

$$F(\rho, |\phi\rangle) = \langle\phi|\rho|\phi\rangle. \tag{1.34}$$

In this thesis the principal reference is $|\Psi^+\rangle$, so that

$$F_{\Psi^+} = \langle\Psi^+|\rho|\Psi^+\rangle. \tag{1.35}$$

Fidelity measures similarity to a chosen target and must therefore not be interpreted directly as an entanglement measure. A local unitary transformation can substantially change F_{Ψ^+} while leaving the amount of entanglement unchanged.

1.6.3. Concurrence and Tangle

For two-qubit systems, concurrence provides a standard quantitative measure of bipartite entanglement.

For a pure state

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle,$$

the concurrence is

$$C(|\psi\rangle) = 2|ad - bc|. \tag{1.36}$$

It satisfies

$$0 \le C \le 1,$$

with $C = 0$ for separable pure states and $C = 1$ for maximally entangled two-qubit states.

For pure bipartite states, concurrence can also be expressed through either reduced density operator,

$$C(|\psi\rangle) = 2\sqrt{\det \rho_A} = \sqrt{2\,[1 - \mathrm{Tr}(\rho_A^2)]}. \tag{1.37}$$

For a mixed two-qubit state, concurrence is defined through the Wootters spin-flip construction [13]. Defining

$$\widetilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y), \tag{1.38}$$

and denoting by $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4$ the square roots of the eigenvalues of $\rho\widetilde{\rho}$, one obtains

$$C(\rho) = \max\left(0,\, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\right). \tag{1.39}$$

The associated tangle is

$$\tau(\rho) = C(\rho)^2. \tag{1.40}$$

For the Bell state,

$$C(|\Psi^+\rangle) = 1, \qquad \tau(|\Psi^+\rangle) = 1.$$

Both quantities are invariant under local unitary transformations.

1.6.4. Entanglement of Formation

The entanglement of formation provides a resource-oriented measure of bipartite entanglement. For two qubits, it can be expressed analytically in terms of the concurrence [13]:

$$E_F(\rho) = h\left(\frac{1 + \sqrt{1 - C(\rho)^2}}{2}\right), \tag{1.41}$$

where

$$h(x) = -x\log_2 x - (1-x)\log_2(1-x) \tag{1.42}$$

is the binary entropy.

For two-qubit states,

$$0 \leq E_F \leq 1,$$

with $E_F = 0$ for separable states and $E_F = 1$ for maximally entangled Bell states.

Because E_F depends nonlinearly on concurrence, it provides information complementary to C and τ, while preserving the same ordering of two-qubit entanglement.

1.7. Bell Inequalities and CHSH Nonlocality

Entanglement does not exhaust the possible manifestations of nonclassical correlations. Bell inequalities provide a stronger test by asking whether the observed correlations can be reproduced by a local hidden-variable theory [14, 15].

In a local hidden-variable description, measurement outcomes on the two arms are represented by functions

$$A(\mathbf{a}, \lambda) \in \{-1, +1\}, \qquad B(\mathbf{b}, \lambda) \in \{-1, +1\},$$

where λ denotes shared hidden information and $\mathbf{a}$, $\mathbf{b}$ are the local measurement settings.

The corresponding correlation function is

$$E(\mathbf{a}, \mathbf{b}) = \int d\lambda\, p(\lambda)\, A(\mathbf{a}, \lambda) B(\mathbf{b}, \lambda). \tag{1.43}$$

The Clauser–Horne–Shimony–Holt form of Bell's inequality considers two measurement settings on each side. The CHSH parameter is

$$S = E(\mathbf{a}_1, \mathbf{b}_1) + E(\mathbf{a}_1, \mathbf{b}_2) + E(\mathbf{a}_2, \mathbf{b}_1) - E(\mathbf{a}_2, \mathbf{b}_2). \tag{1.44}$$

Any local hidden-variable theory satisfies

$$|S| \leq 2. \tag{1.45}$$

Quantum mechanics allows this bound to be exceeded, while respecting the Tsirelson bound

$$|S| \leq 2\sqrt{2}. \tag{1.46}$$

Using the correlation tensor, Eq. (1.44) becomes

$$S = \mathbf{a}_1^\mathsf{T} T \mathbf{b}_1 + \mathbf{a}_1^\mathsf{T} T \mathbf{b}_2 + \mathbf{a}_2^\mathsf{T} T \mathbf{b}_1 - \mathbf{a}_2^\mathsf{T} T \mathbf{b}_2. \tag{1.47}$$

For a given two-qubit state, the maximal CHSH value can be obtained directly from the tensor [16]. Defining

$$M = T^\mathsf{T} T, \tag{1.48}$$

and denoting by m_1 and m_2 the two largest eigenvalues of M, one has

$$S_{\max} = 2\sqrt{m_1 + m_2}. \tag{1.49}$$

The distinction between S and $S_{\max}$ is particularly important in this work. The first quantity corresponds to a fixed set of experimental measurement axes and can change when the state undergoes a coherent polarization rotation. The second is optimized over all local measurement directions and characterizes the maximum CHSH violation supported by the state itself.

Entanglement and CHSH nonlocality are therefore related but distinct properties. In particular, an entangled mixed state need not violate the CHSH inequality.

1.8. Complementarity of the State Metrics

The quantities introduced in the previous sections characterize different properties of the same density operator and should therefore be interpreted jointly.

Purity quantifies whether the state is pure or mixed but does not measure entanglement. Fidelity measures the overlap with a selected reference state and can change under coherent local transformations. Concurrence, tangle, and entanglement of formation quantify bipartite entanglement, whereas the CHSH parameter tests whether the observed correlations are incompatible with a local hidden-variable description.

This distinction is particularly relevant for optical-fiber propagation. A coherent polarization transformation may alter the correlations measured along fixed laboratory axes and reduce fidelity with respect to the initial Bell state without changing purity or entanglement. By contrast, unresolved statistical fluctuations or coupling between polarization and unobserved degrees of freedom may produce genuine mixedness and a reduction of concurrence.

The simultaneous evaluation of several metrics therefore makes it possible to distinguish changes caused by coherent state rotation from those associated with decoherence and entanglement degradation.

These concepts provide the mathematical basis for Chapter 2, where progressively more detailed models of optical-fiber propagation are introduced and their effects on the quantities defined here are derived.

2 | Fiber-Channel Modeling

The mathematical framework introduced in Chapter 1 provides the tools required to describe bipartite polarization states, local unitary evolution, quantum channels, polarization correlations, and entanglement metrics. The purpose of the present chapter is to apply those tools to the progressive construction of models for the propagation of polarization-entangled photon pairs through optical fibers.

The modeling strategy proceeds from analytically controlled effective descriptions toward a frequency-dependent representation of the underlying fiber physics. The first models retain only the polarization degree of freedom and isolate coherent phase evolution, statistical phase fluctuations, and depolarization. These mechanisms are subsequently combined into a two-arm effective fiber channel.

The monochromatic approximation is then relaxed by explicitly introducing the spectral structure of the photons and the frequency dependence of birefringent fiber propagation. This leads to a polarization-mode-dispersion (PMD) model in which the complete polarization–frequency evolution remains unitary for an ideal lossless fiber, while an effectively mixed polarization state can emerge after unresolved spectral and temporal degrees of freedom are traced out.

Figure 2.1 summarizes the progressive hierarchy adopted throughout the chapter.

2.1. Reduced Phase Model

A general lossless optical fiber may produce an arbitrary unitary transformation of the polarization state. The first propagation model considered in this work is a reduced description of this general evolution and is motivated by the polarization-compensation procedure employed in the experimental setup.

Polarization compensation is used to realign the dominant output polarization basis with the reference basis of the measurement system. Such compensation does not, however, imply a complete reconstruction or inversion of the full fiber Jones operator. In particular, a residual rotation around the stabilized polarization axis may remain.

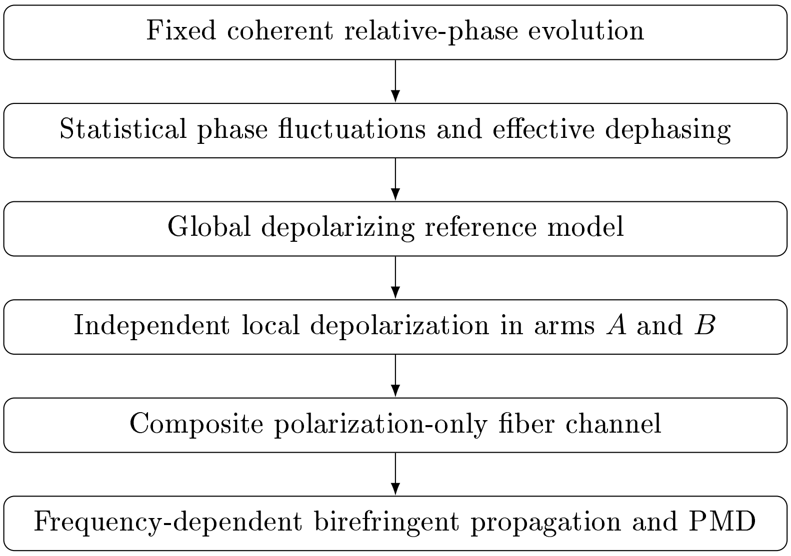

Figure 2.1: Modeling hierarchy adopted in this work. The first levels provide analytically controlled polarization-only descriptions, whereas the final level explicitly retains the spectral dependence of birefringent propagation.

Within the computational basis adopted in this thesis, this residual coherent effect is represented by a relative phase between the horizontal and vertical polarization components. The general polarization transformation is therefore reduced to a single physically relevant phase parameter.

For one polarization qubit, the corresponding unitary operator is

$$U(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \tag{2.1}$$

defined up to an irrelevant global phase. Equivalently, this transformation corresponds, up to a global phase factor, to a rotation around the z-axis of the Bloch sphere.

At this modeling level, the phase is assumed to be fixed for each propagation realization. Statistical fluctuations of the same reduced phase parameter are introduced separately in the following model.

If the two fiber arms introduce residual phases θ_A and θ_B, respectively, their joint action is

$$U_A(\theta_A) \otimes U_B(\theta_B). \tag{2.2}$$

Applying this transformation to the reference Bell state gives

$$|\psi(\theta_A,\theta_B)\rangle = \frac{1}{\sqrt{2}}\left(e^{i\theta_B}|01\rangle + e^{i\theta_A}|10\rangle\right). \tag{2.3}$$

Factoring out the global phase $e^{i\theta_B}$ yields

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + e^{i\theta}|10\rangle\right), \qquad \theta = \theta_A - \theta_B. \tag{2.4}$$

Only the relative phase between the two arms therefore affects polarization observables. The two independent local phase parameters can consequently be reduced to the single effective parameter θ.

With the convention adopted throughout this work, the same state may be generated by assigning the complete relative transformation to arm A,

$$U_\theta^{red} = U_A(\theta) \otimes I_B. \tag{2.5}$$

This convention simplifies the analytical and numerical representation without changing any observable prediction. The physical quantity represented by θ remains the differential phase accumulated between the two propagation arms.

The term *reduced phase model* will be used throughout this work for this one-parameter effective description. Its relation to the more general fixed-frequency birefringent transformation is derived explicitly in Section 2.8.1.

2.1.1. Correlation Structure

For the state in Eq. (2.4), the correlation tensor defined in Chapter 1 is

$$T(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{2.6}$$

The phase therefore rotates the transverse correlation structure in the xy-plane while leaving the longitudinal anticorrelation unchanged:

$$T_{xx} = T_{yy} = \cos\theta, \qquad T_{xy} = -\sin\theta, \qquad T_{yx} = \sin\theta, \qquad T_{zz} = -1. \tag{2.7}$$

The transformation is local and unitary. Consequently,

$$\gamma = 1, \qquad C = 1, \qquad \tau = 1, \qquad E_F = 1 \tag{2.8}$$

for every value of θ.

The fidelity with respect to the original Bell state is instead

$$F_{\Psi^+} = \frac{1 + \cos\theta}{2}. \tag{2.9}$$

Thus, a reduction in fixed-reference Bell-state fidelity does not by itself imply entanglement degradation. It may result entirely from a coherent change in the orientation of the state relative to the selected reference basis.

The same distinction appears in Bell tests. A CHSH parameter evaluated using fixed laboratory axes may vary with θ, whereas optimization over all local measurement directions gives

$$S_{\max} = 2\sqrt{2}, \tag{2.10}$$

independently of the phase.

The reduced phase model therefore establishes the first important reference case of this thesis: coherent changes in fixed-basis measurements must be distinguished from genuine decoherence and entanglement degradation.

2.2. Statistical Phase Fluctuations and Dephasing

The fixed-phase description assumes that the relative phase is known and remains constant over the relevant observation interval. In practice, birefringence and optical-path conditions may vary between successive photon-pair realizations. If these fluctuations are not resolved by the measurement system, the observed state must be obtained by averaging over the corresponding phase realizations.

Let p_k denote the probability associated with the phase θ_k. The ensemble density operator is

$$\rho_{AB}^{\text{ens}} = \sum_k p_k |\psi(\theta_k)\rangle\langle\psi(\theta_k)|, \qquad \sum_k p_k = 1. \tag{2.11}$$

The phase statistics can be summarized through the complex coherence parameter

$$\mu = \left\langle e^{i\theta} \right\rangle = \sum_k p_k e^{i\theta_k} = \langle\cos\theta\rangle + i\langle\sin\theta\rangle. \tag{2.12}$$

For a continuous probability density $P(\theta)$,

$$\mu = \int d\theta \, P(\theta) e^{i\theta}. \tag{2.13}$$

It is useful to write

$$\mu = |\mu| e^{i\phi_\mu}. \tag{2.14}$$

The magnitude $|\mu|$ quantifies the coherence surviving the statistical averaging, while the argument ϕ_μ represents the residual coherent orientation of the state in the transverse polarization plane.

In the computational basis, the resulting state is

$$\rho_{AB}^{\text{ens}} = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & \mu^* & 0 \\ 0 & \mu & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{2.15}$$

The diagonal populations remain unchanged, whereas the off-diagonal coherence terms are multiplied by μ. The model therefore corresponds to a random-unitary dephasing process together with a possible residual coherent phase rotation when $\arg(\mu) \neq 0$.

2.2.1. Correlations and State Metrics

The correlation tensor becomes

$$T_\mu = \begin{pmatrix} \mathrm{Re}(\mu) & -\mathrm{Im}(\mu) & 0 \\ \mathrm{Im}(\mu) & \mathrm{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{2.16}$$

The magnitude $|\mu|$ therefore determines the surviving transverse correlation amplitude, whereas the argument of μ determines its remaining orientation.

For this family of states,

$$\gamma = \frac{1 + |\mu|^2}{2}, \tag{2.17}$$

$$F_{\Psi^+} = \frac{1 + \mathrm{Re}(\mu)}{2}, \tag{2.18}$$

and

$$C = |\mu|, \qquad \tau = |\mu|^2. \tag{2.19}$$

The corresponding entanglement of formation is

$$E_F = h\left(\frac{1 + \sqrt{1 - |\mu|^2}}{2}\right), \tag{2.20}$$

where $h(x)$ is the binary entropy introduced in Chapter 1.

The maximal CHSH value is

$$S_{\max} = 2\sqrt{1 + |\mu|^2}. \tag{2.21}$$

The limiting cases provide a direct physical interpretation. For a deterministic phase,

$$|\mu| = 1, \tag{2.22}$$

and the state remains pure and maximally entangled. For a phase uniformly distributed over $[0, 2\pi]$,

$$\mu = 0, \tag{2.23}$$

and the transverse coherences vanish completely. The resulting state is

$$\rho_{\mu=0} = \frac{1}{2}\left(|01\rangle\langle 01| + |10\rangle\langle 10|\right), \tag{2.24}$$

which retains perfect classical anticorrelation in the computational basis but is no longer entangled.

Intermediate phase distributions produce partially coherent mixed states.

This model represents decoherence at the reduced polarization level, but it does not yet specify the microscopic physical mechanism responsible for the unresolved phase distribution. A frequency-resolved mechanism producing related losses of polarization coherence is introduced later in this chapter.

2.3. Depolarizing Reference Models

Statistical phase averaging suppresses specific coherence terms while preserving the longitudinal population correlation. A complementary analytical reference is provided by isotropic depolarization, which suppresses polarization information uniformly in all directions.

The depolarizing models introduced here are phenomenological. They do not represent a microscopic model of PMD. Their purpose is to provide analytically controlled descriptions of polarization degradation before the physical origin of decoherence is treated explicitly through frequency-dependent propagation.

2.3.1. Global Depolarizing Baseline

For a two-qubit density operator, the global depolarizing channel is defined as

$$\mathcal{D}_p^4(\rho) = (1-p)\rho + p\frac{I_4}{4}, \qquad 0 \le p \le 1. \tag{2.25}$$

For a Bell-state input,

$$\rho(p) = (1-p)|\Psi^+\rangle\langle\Psi^+| + p\frac{I_4}{4}. \tag{2.26}$$

The corresponding purity is

$$\gamma = 1 - \frac{3}{2}p + \frac{3}{4}p^2, \tag{2.27}$$

the Bell-state fidelity is

$$F_{\Psi^+} = 1 - \frac{3}{4}p, \tag{2.28}$$

and the concurrence is

$$C = \max\left(0,\, 1 - \frac{3}{2}p\right). \tag{2.29}$$

All bipartite Pauli correlations are contracted according to

$$T_{\text{out}} = (1-p)T_{\text{in}}. \tag{2.30}$$

For a Bell-state input,

$$S_{\max} = 2\sqrt{2}(1-p). \tag{2.31}$$

The global channel is useful as a validation benchmark, but it is not directly aligned with the geometry of the physical system considered in this thesis, where the two photons propagate through separate fibers.

2.3.2. Local Two-Arm Depolarization

A physically more appropriate effective description assigns an independent depolarizing channel to each propagation arm.

The parameter convention adopted throughout this work is defined directly by the Bloch-vector contraction. A single-qubit depolarizing channel is written as

$$\mathcal{D}_p(\rho) = (1-p)\rho + p\frac{I_2}{2}, \tag{2.32}$$

so that $p = 0$ corresponds to no depolarization and $p = 1$ to complete depolarization.

Its linear extension to a generic operator X is

$$\mathcal{D}_p(X) = (1-p)X + p\frac{I_2}{2}\mathrm{Tr}(X). \tag{2.33}$$

For a single-qubit state written in Bloch form,

$$\rho = \frac{1}{2}\left(I + \mathbf{r}\cdot\boldsymbol{\sigma}\right), \tag{2.34}$$

the channel gives

$$\mathcal{D}_p(\rho) = \frac{1}{2}\left[I + (1-p)\mathbf{r}\cdot\boldsymbol{\sigma}\right]. \tag{2.35}$$

The Bloch vector therefore undergoes the isotropic contraction

$$\mathbf{r} \longmapsto (1-p)\mathbf{r}. \tag{2.36}$$

For two independent fiber arms,

$$\rho_{AB}^{\mathrm{out}} = \left(\mathcal{D}_{p_A}\otimes\mathcal{D}_{p_B}\right)\left(\rho_{AB}^{\mathrm{in}}\right). \tag{2.37}$$

Although the effective noise acts locally, the channel acts on the complete bipartite density operator. The entanglement and two-photon correlations are properties of the joint state and cannot be reconstructed from the two reduced states alone.

2.3.3. Correlation-Survival Factor

The single-qubit depolarizing channel acts on the Pauli basis as

$$\mathcal{D}_p(I) = I, \qquad \mathcal{D}_p(\sigma_i) = (1-p)\sigma_i, \qquad i \in \{x,y,z\}. \tag{2.38}$$

Consequently, each two-body Pauli correlation is multiplied by the factor

$$\eta = (1-p_A)(1-p_B). \tag{2.39}$$

The correlation tensor therefore satisfies

$$T_{\text{out}} = \eta T_{\text{in}}, \tag{2.40}$$

and, for arbitrary local measurement axes,

$$E_{\text{out}}(\mathbf{a}, \mathbf{b}) = \eta E_{\text{in}}(\mathbf{a}, \mathbf{b}). \tag{2.41}$$

The factor η will be used throughout the remainder of the effective fiber-channel models as the survival factor of bipartite polarization correlations.

2.4. Composite Effective Fiber Channel

The fixed relative-phase model and the local depolarizing model represent two different classes of propagation effects. The former is coherent and rotates the correlation structure, whereas the latter is non-unitary at the polarization level and suppresses its magnitude.

They are combined through the effective channel

$$\rho_{AB}^{\text{out}} = (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_\theta \rho_{AB}^{\text{in}} U_\theta^\dagger \right]. \tag{2.42}$$

For isotropic depolarization, the local depolarizing channel is covariant under single-qubit unitary transformations. Consequently, for the phase transformation considered here,

$$(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_\theta \rho U_\theta^\dagger \right] = U_\theta \left[(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) (\rho) \right] U_\theta^\dagger. \tag{2.43}$$

Thus, within this isotropic model, the order of phase evolution and depolarization does not affect the final state. This property is specific to the adopted channel and does not hold for arbitrary anisotropic noise mechanisms.

2.4.1. Fixed-Phase Composite Model

For the maximally entangled input family considered here, the reduced states of both subsystems remain $I_2/2$. The action of the two local depolarizing channels can therefore be written compactly as

$$\rho_{AB}^{\text{out}} = \eta |\psi(\theta)\rangle\langle\psi(\theta)| + (1 - \eta) \frac{I_4}{4}. \tag{2.44}$$

The corresponding correlation tensor is

$$T_{\mathrm{out}} = \eta \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{2.45}$$

The parameter θ therefore controls the orientation of the transverse correlations, whereas η controls their magnitude.

The corresponding purity is

$$\gamma = \frac{1+3\eta^2}{4}, \tag{2.46}$$

while the Bell-state fidelity is

$$F_{\Psi^+} = \frac{1-\eta}{4} + \eta\frac{1+\cos\theta}{2}. \tag{2.47}$$

The concurrence is

$$C = \max\left(0, \frac{3\eta-1}{2}\right), \tag{2.48}$$

and the maximal CHSH parameter is

$$S_{\max} = 2\sqrt{2}\,\eta. \tag{2.49}$$

Entanglement disappears when

$$\eta \leq \frac{1}{3}, \tag{2.50}$$

whereas CHSH violation requires

$$\eta > \frac{1}{\sqrt{2}}. \tag{2.51}$$

The separation between these thresholds illustrates that entanglement can survive after CHSH nonlocality is no longer observable.

2.4.2. Statistical Composite Model

The composite description can be extended by allowing the relative phase to fluctuate between realizations. The ensemble-averaged state is

$$\rho_{AB}^{\mathrm{ens}} = \sum_k p_k \left(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}\right) \left[U_{\theta_k} \rho_{AB}^{\mathrm{in}} U_{\theta_k}^\dagger\right]. \tag{2.52}$$

Using the covariance property in Eq. (2.43),

$$\rho_{AB}^{\mathrm{ens}} = \left(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}\right) \left[\sum_k p_k U_{\theta_k} \rho_{AB}^{\mathrm{in}} U_{\theta_k}^\dagger\right]. \tag{2.53}$$

For the random-phase family of Section 2.2, this reduces to

$$\rho_{AB}^{\mathrm{ens}} = \eta \rho_{AB}^{\mu} + (1-\eta)\frac{I_4}{4}, \tag{2.54}$$

where ρ_{AB}^{μ} denotes the state in Eq. (2.15).

The output correlation tensor is

$$T_{\mathrm{out}}^{\mathrm{ens}} = \eta \begin{pmatrix} \mathrm{Re}(\mu) & -\mathrm{Im}(\mu) & 0 \\ \mathrm{Im}(\mu) & \mathrm{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{2.55}$$

The two mechanisms therefore leave distinguishable signatures. Phase averaging suppresses the transverse coherence through μ, whereas local isotropic depolarization multiplies all bipartite correlations by η.

The purity becomes

$$\gamma = \frac{1 + \eta^2\left(1 + 2|\mu|^2\right)}{4}, \tag{2.56}$$

the fidelity is

$$F_{\Psi^+} = \frac{1-\eta}{4} + \eta\frac{1 + \mathrm{Re}(\mu)}{2}, \tag{2.57}$$

and the concurrence is

$$C = \max\left(0,\ \eta|\mu| - \frac{1-\eta}{2}\right). \tag{2.58}$$

The maximal CHSH parameter is

$$S_{\max} = 2\eta\sqrt{1+|\mu|^2}. \tag{2.59}$$

Bell-inequality violation therefore requires

$$\eta\sqrt{1+|\mu|^2} > 1. \tag{2.60}$$

At the level of transverse correlations, the progression can therefore be summarized schematically as

$$e^{i\theta} \quad \longrightarrow \quad \mu = \langle e^{i\theta} \rangle \quad \longrightarrow \quad \eta\mu.$$

The effective model is analytically transparent and useful for distinguishing coherent phase evolution, statistical dephasing, and isotropic polarization degradation. Its main limitation is that the parameters μ, p_A, and p_B describe reduced behavior rather than being derived directly from the spectral and birefringent properties of the physical fibers.

2.5. From Effective Channels to Frequency-Dependent Propagation

The effective models introduced above treat each photon exclusively as a polarization qubit and assume that the relevant fiber action can be represented without explicitly retaining optical frequency.

This approximation is appropriate when the polarization transformation varies negligibly across the photon bandwidth, or when the unresolved physical mechanisms can be represented adequately by phenomenological channel parameters.

The effective models nevertheless omit several properties of real birefringent fiber propagation. In particular, they contain no explicit optical bandwidth, principal states of polarization, differential group delay, or frequency-dependent polarization transformation. Moreover, isotropic depolarization cannot reproduce the directional character of PMD: a state aligned with a principal state may remain fully polarized, whereas a superposition of the principal states may become correlated with arrival time and lose reduced polarization coherence.

The polarization-only models also do not distinguish between two physically different origins of mixedness:

- averaging over different propagation realizations occurring at different times;
- polarization becoming correlated with unresolved spectral or temporal degrees of freedom within an individual photon wave packet.

Both mechanisms can produce mixed reduced polarization states, but their physical interpretations are different.

SPDC photons have finite optical bandwidth. Their spectral structure is determined by the pump envelope and by the phase-matching properties of the nonlinear source [17]. Different frequency components may therefore experience different birefringent transformations during propagation.

The local polarization operator must consequently be generalized to

$$U_K \longrightarrow U_K(\omega_K), \qquad K \in \{A, B\}. \tag{2.61}$$

For a resolved pair of frequencies,

$$U_{AB}(\omega_A, \omega_B) = U_A(\omega_A) \otimes U_B(\omega_B). \tag{2.62}$$

The complete polarization–frequency evolution of an ideal lossless fiber remains unitary. Nevertheless, frequency-dependent propagation can correlate polarization with frequency and arrival time. If these degrees of freedom are not resolved by the measurement system, the observable polarization state can become mixed.

This provides the physical connection with the phenomenological models developed above: an effective non-unitary polarization channel can emerge from an underlying unitary evolution when only part of the complete optical system is observed.

2.5.1. Illustrative Effect of Frequency-Dependent Birefringence

The physical consequence of the frequency dependence introduced in Eq. (2.61) can be understood qualitatively by considering different spectral components of the same photon.

In a birefringent medium, the polarization transformation generally depends on optical frequency. Consequently, two frequency components ω_1 and ω_2 need not undergo the same transformation,

$$U(\omega_1) \neq U(\omega_2). \tag{2.63}$$

For example, if the input polarization contains components along both local birefringent eigenmodes, the two components accumulate a relative phase during propagation. Because the birefringence is generally dispersive, this relative phase depends on frequency. Different parts of the photon spectrum can therefore emerge from the fiber with different polarization states.

For a sufficiently narrow optical spectrum, the transformation may be considered approximately constant over the occupied bandwidth,

$$U(\omega) \simeq U(\omega_0), \tag{2.64}$$

where ω_0 denotes the central optical frequency. In this limit, all spectral components undergo essentially the same polarization transformation and the polarization dynamics can be described accurately by a single-frequency unitary operator.

When the frequency dependence cannot be neglected, however, different spectral components experience different polarization transformations. Polarization then becomes correlated with the spectral degree of freedom and, equivalently in the time domain, with the temporal structure of the photon wave packet.

The complete optical evolution can still be unitary and preserve the purity of the full state. Nevertheless, a detector that does not resolve the spectral or temporal degree of freedom has access only to the polarization subsystem. The corresponding reduced polarization state may therefore exhibit a loss of coherence and become mixed.

This mechanism is fundamentally different from the phenomenological isotropic depolarizing channel introduced in Section 2.3. Here, the apparent polarization decoherence originates from coherent frequency-dependent propagation followed by the neglect of an unresolved degree of freedom.

The spectral Hilbert space, the finite-bandwidth two-photon state, and the corresponding reduction to an effective polarization state are introduced formally in the following section. The relation between frequency-dependent birefringence, differential phase, differential group delay, and PMD is then developed subsequently.

2.6. Spectral Description of SPDC Photon Pairs

The polarization-only description introduced previously treats each photon as a two-level system,

$$\mathcal{H}_{\mathrm{pol},K} \simeq \mathbb{C}^2, \qquad K \in \{A, B\}. \tag{2.65}$$

This description is sufficient as long as the optical transformation can be regarded as independent of frequency. Polarization-mode dispersion, however, originates precisely from a frequency-dependent polarization transformation. The spectral degree of freedom must therefore be included explicitly in the state space [5, 7, 10].

For each photon K, the Hilbert space is consequently extended to

$$\mathcal{H}_K = \mathcal{H}_{\omega_K} \otimes \mathcal{H}_{\mathrm{pol},K}, \qquad K \in \{A, B\}, \tag{2.66}$$

where $\mathcal{H}_{\omega_K}$ describes the continuous spectral degree of freedom and $\mathcal{H}_{\mathrm{pol},K}$ describes the polarization qubit. The complete two-photon space is therefore

$$\mathcal{H}_{\mathrm{tot}} = \mathcal{H}_A \otimes \mathcal{H}_B = (\mathcal{H}_{\omega_A} \otimes \mathcal{H}_{\mathrm{pol},A}) \otimes (\mathcal{H}_{\omega_B} \otimes \mathcal{H}_{\mathrm{pol},B})\,. \tag{2.67}$$

For convenience, the tensor factors are reordered throughout this work so that the two spectral degrees of freedom and the two polarization degrees of freedom are grouped together:

$$\mathcal{H}_{\mathrm{tot}} = \underbrace{(\mathcal{H}_{\omega_A} \otimes \mathcal{H}_{\omega_B})}_{\mathcal{H}_{\omega,\mathrm{tot}}} \otimes \underbrace{(\mathcal{H}_{\mathrm{pol},A} \otimes \mathcal{H}_{\mathrm{pol},B})}_{\mathcal{H}_{\mathrm{pol,tot}}}\,. \tag{2.68}$$

The two orderings are physically equivalent; the second one is adopted because it makes the separation between spectral and polarization degrees of freedom explicit and allows the spectral partial trace introduced later to be written naturally.

Frequency eigenstates are denoted by $|\omega_K\rangle$. Since frequency is a continuous variable, these states obey Dirac-delta normalization,

$$\langle \omega_K | \omega'_K \rangle = \delta(\omega_K - \omega'_K), \qquad K \in \{A, B\}, \tag{2.69}$$

together with the completeness relation

$$\int d\omega_K \, |\omega_K\rangle\langle\omega_K| = I_{\omega_K}. \tag{2.70}$$

The corresponding two-photon frequency basis states are

$$|\omega_A, \omega_B\rangle = |\omega_A\rangle \otimes |\omega_B\rangle, \tag{2.71}$$

and satisfy

$$\langle \omega_A, \omega_B | \omega'_A, \omega'_B \rangle = \delta(\omega_A - \omega'_A)\delta(\omega_B - \omega'_B). \tag{2.72}$$

These relations are the continuous-variable counterpart of the orthogonality and completeness relations used for discrete quantum bases.

The finite-bandwidth spectral state generated by the SPDC source can then be expanded in this basis as

$$|F\rangle_{\omega_A\omega_B} = \iint d\omega_A \, d\omega_B \, F(\omega_A, \omega_B)|\omega_A, \omega_B\rangle, \tag{2.73}$$

where

$$F(\omega_A, \omega_B) = \langle \omega_A, \omega_B | F \rangle \tag{2.74}$$

is the joint spectral amplitude (JSA) of the photon pair. It therefore plays the role of a two-photon wavefunction in the frequency representation: for each pair (ω_A, ω_B), it specifies the complex probability amplitude associated with generating photon A at frequency ω_A and photon B at frequency ω_B [17].

The spectral state is normalized according to

$$\iint d\omega_A \, d\omega_B \, |F(\omega_A, \omega_B)|^2 = 1. \tag{2.75}$$

The corresponding joint spectral intensity (JSI) is

$$J(\omega_A, \omega_B) = |F(\omega_A, \omega_B)|^2, \tag{2.76}$$

which represents the joint probability density of the two photon frequencies. In particular,

$$dP = J(\omega_A, \omega_B)\, d\omega_A\, d\omega_B \tag{2.77}$$

is the probability of finding the two photons within infinitesimal spectral intervals around ω_A and ω_B, respectively.

The corresponding marginal spectra are obtained by integrating over the frequency of the other photon,

$$P_A(\omega_A) = \int d\omega_B\, J(\omega_A, \omega_B), \tag{2.78}$$

$$P_B(\omega_B) = \int d\omega_A\, J(\omega_A, \omega_B). \tag{2.79}$$

The joint spectral structure originates from the physical SPDC generation process. In spontaneous parametric down-conversion, one pump photon is converted into two lower-frequency photons conventionally referred to as signal and idler photons. In the two-arm notation adopted here, these photons are associated with subsystems A and B [17].

Energy conservation imposes

$$\omega_p = \omega_A + \omega_B. \tag{2.80}$$

For an ideal monochromatic pump of angular frequency $\omega_{p,0}$, this constraint fixes the generated frequencies to the line

$$\omega_A + \omega_B = \omega_{p,0}. \tag{2.81}$$

A realistic pump instead possesses a finite spectral bandwidth. Its spectral amplitude is described by the pump envelope $\alpha_p(\omega_p)$, which, after imposing energy conservation, enters the two-photon state as

$$\alpha_p(\omega_A + \omega_B). \tag{2.82}$$

The generated frequencies must also satisfy the phase-matching condition of the nonlinear medium. Introducing the longitudinal wave-vector mismatch

$$\Delta k(\omega_A, \omega_B) = k_p(\omega_A + \omega_B) - k_A(\omega_A) - k_B(\omega_B), \tag{2.83}$$

a commonly used form for a uniform nonlinear interaction of length L is

$$\Phi(\omega_A, \omega_B) = \operatorname{sinc}\left[\frac{\Delta k(\omega_A, \omega_B)L}{2}\right] \exp\left[i\frac{\Delta k(\omega_A, \omega_B)L}{2}\right], \tag{2.84}$$

up to convention-dependent phase factors and possible additional terms associated with the specific nonlinear-crystal configuration [17].

The joint spectral amplitude can therefore be written schematically as

$$F(\omega_A, \omega_B) = \mathcal{N}\, \alpha_p(\omega_A + \omega_B)\Phi(\omega_A, \omega_B), \tag{2.85}$$

where $\mathcal{N}$ is chosen such that Eq. (2.75) is satisfied [17, 18].

Equation (2.85) separates two physically different constraints. The pump envelope determines which values of the sum frequency $\omega_A + \omega_B$ are supplied by the pump, whereas $\Phi(\omega_A, \omega_B)$ determines which of these frequency pairs are efficiently generated by the phase-matching properties of the nonlinear medium. The observed joint spectrum results from the overlap of these two spectral constraints.

Consequently, the generated frequencies need not be statistically independent. For example, a narrowband pump strongly constrains $\omega_A + \omega_B$, so that an increase in one photon frequency must be accompanied by a corresponding decrease in the other. This produces the frequency anticorrelation commonly encountered in SPDC photon pairs [17].

More generally, complete spectral separability requires the joint spectral amplitude to factorize as

$$F(\omega_A, \omega_B) = f_A(\omega_A) f_B(\omega_B). \tag{2.86}$$

When such a decomposition is not possible, the spectral state cannot be written as a product of two independent single-photon spectral states. This distinction is important because the phase of the JSA, and not only its intensity $J = |F|^2$, can contribute to spectral correlations.

2.6.1. Input Polarization–Frequency State

The propagation model developed in this work introduces an additional assumption concerning the relation between the spectral and polarization degrees of freedom at the source output. These degrees of freedom are assumed to be initially separable, so that the polarization state is the same for every generated frequency pair.

In particular, the two polarization alternatives forming the Bell state are assumed to possess the same joint spectral amplitude. Under this condition, the multimode two-photon state can be factorized into independent spectral and polarization components [19].

Physically, this assumption corresponds to neglecting source-induced polarization–frequency distinguishability, or equivalently to assuming that such distinguishability has been sufficiently suppressed through source design, compensation, and spectral filtering. In practical polarization-entangled SPDC sources, spectral differences between alternative generation processes can otherwise provide distinguishing information and reduce the observed polarization entanglement [20].

A factorized temporal–polarization input state is also commonly adopted in theoretical treatments of PMD, where coupling between these degrees of freedom is subsequently generated by propagation [7].

For the reference Bell state,

$$|\Psi_{\text{tot}}^{\text{in}}\rangle = |F\rangle_{\omega_A\omega_B} \otimes |\Psi^+\rangle_{\text{pol}}. \tag{2.87}$$

Equivalently, using the spectral expansion introduced in Eq. (2.73),

$$|\Psi_{\text{tot}}^{\text{in}}\rangle = \iint d\omega_A\, d\omega_B\, F(\omega_A,\omega_B)|\omega_A,\omega_B\rangle \otimes |\Psi^+\rangle_{\text{pol}}. \tag{2.88}$$

More generally, allowing the polarization state itself to be mixed,

$$\rho_{\text{tot}}^{\text{in}} = |F\rangle\langle F| \otimes \rho_{AB}^{\text{pol,in}}. \tag{2.89}$$

For the reference pure Bell-state input,

$$\rho_{AB}^{\mathrm{pol,in}} = |\Psi^+\rangle\langle\Psi^+|. \tag{2.90}$$

This factorization is an explicit modeling assumption rather than a general property of type-II SPDC sources. It states that the two polarization alternatives forming the Bell state share the same joint spectral state, so that the source initially carries no spectral information capable of distinguishing them.

Initial polarization–frequency correlations are therefore neglected. Within the present model, any such coupling appearing after transmission can consequently be attributed to the frequency dependence of the fiber propagation [5, 7].

This factorized state provides the starting point for the PMD model developed in the following sections. Once the local fiber transformations become frequency-dependent, different frequency pairs generally undergo different polarization transformations, and the factorized structure of the input state need not be preserved.

2.6.2. Frequency-Controlled Fiber Evolution

An ideal stationary lossless fiber preserves optical frequency while applying a frequency-dependent polarization transformation. In the tensor ordering of Eq. (2.68), the complete two-arm propagation operator can therefore be written directly as

$$\mathcal{U}_{AB} = \iint d\omega_A \, d\omega_B \, |\omega_A, \omega_B\rangle\langle\omega_A, \omega_B| \otimes [U_A(\omega_A) \otimes U_B(\omega_B)] \, . \tag{2.91}$$

The operator in Eq. (2.91) is diagonal in the frequency basis. The projector $|\omega_A, \omega_B\rangle\langle\omega_A, \omega_B|$ does not represent a physical measurement of frequency. Rather, it ensures that each spectral component of the state is associated with the polarization transformation corresponding to that particular frequency pair. The frequencies themselves are left unchanged, while the polarization transformation depends on their values.

This action can be made explicit by considering first a state with a definite frequency pair (ω'_A, ω'_B) and an arbitrary two-photon polarization state $|\chi\rangle$,

$$|\Psi\rangle = |\omega'_A, \omega'_B\rangle \otimes |\chi\rangle. \tag{2.92}$$

Applying Eq. (2.91) gives

$$\mathcal{U}_{AB}|\Psi\rangle = \iint d\omega_A\, d\omega_B\, |\omega_A, \omega_B\rangle\langle\omega_A, \omega_B|\omega'_A, \omega'_B\rangle \otimes [U_A(\omega_A) \otimes U_B(\omega_B)]\, |\chi\rangle. \tag{2.93}$$

Using the orthogonality relation

$$\langle\omega_A, \omega_B|\omega'_A, \omega'_B\rangle = \delta(\omega_A - \omega'_A)\delta(\omega_B - \omega'_B), \tag{2.94}$$

the two integrals collapse, yielding

$$\mathcal{U}_{AB}\left(|\omega'_A, \omega'_B\rangle \otimes |\chi\rangle\right) = |\omega'_A, \omega'_B\rangle \otimes [U_A(\omega'_A) \otimes U_B(\omega'_B)]\, |\chi\rangle. \tag{2.95}$$

Equation (2.95) provides the direct physical interpretation of the full propagation operator. The fiber does not convert the photons from one optical frequency to another. Instead, the frequency pair determines which local polarization transformations are experienced by the two photons. In this sense, Eq. (2.91) can be interpreted as a continuous frequency-controlled polarization transformation.

For a finite-bandwidth input state, many frequency pairs are present simultaneously. Each spectral component therefore undergoes, in general, a different polarization transformation. This is the mechanism through which frequency-dependent propagation can establish correlations between spectral and polarization degrees of freedom.

After propagation,

$$\rho_{\mathrm{tot}}^{\mathrm{out}} = \mathcal{U}_{AB}\rho_{\mathrm{tot}}^{\mathrm{in}}\mathcal{U}_{AB}^{\dagger}. \tag{2.96}$$

If the spectral degrees of freedom are not resolved, the observable polarization state is obtained by tracing them out:

$$\rho_{AB}^{\mathrm{pol,out}} = \mathrm{Tr}_{\omega_A,\omega_B}\left[\rho_{\mathrm{tot}}^{\mathrm{out}}\right]. \tag{2.97}$$

Using the orthogonality of the frequency eigenstates and performing the partial trace over the unresolved spectral degrees of freedom gives [5, 9]

$$\rho_{AB}^{\mathrm{pol,out}} = \iint d\omega_A\, d\omega_B\, J(\omega_A, \omega_B)\, [U_A(\omega_A) \otimes U_B(\omega_B)]\, \rho_{AB}^{\mathrm{pol,in}}\, [U_A(\omega_A) \otimes U_B(\omega_B)]^{\dagger}. \tag{2.98}$$

Equation (2.98) is the central connection between the frequency-dependent physical propagation model and the effective polarization channel.

Only the joint spectral intensity $J = |F|^2$ appears in this expression. Under the assumptions made above, the spectral phase of the JSA cancels after the frequency-preserving propagation and complete spectral trace. This simplification relies on the initial polarization–frequency factorization and on the absence of frequency conversion in the fiber.

The result can be interpreted as a continuous random-unitary channel on the polarization subsystem: each frequency pair produces a local unitary transformation, while the unresolved spectral distribution determines the statistical weighting [5, 10].

2.6.3. Nature of the Reduced Polarization Decoherence

The reduced evolution in Eq. (2.98) should not in general be identified with a standard isotropic depolarizing channel. Its specific form depends on the frequency dependence of the local polarization transformations.

If the birefringence eigenaxes remain fixed while only the differential phase varies with frequency, tracing over the unresolved spectrum produces a dephasing-type channel in that eigenpolarization basis: the corresponding populations are preserved while the off-diagonal coherence terms are reduced.

More general frequency-dependent birefringent propagation may also involve a variation of the effective polarization rotation axes across the spectrum. In this case, the spectral average can contract the polarization state along different Bloch-sphere directions and produce a more general anisotropic polarization decoherence.

Such a reduction may be described as depolarization in the optical sense, since the degree of polarization and the reduced-state purity can decrease, but it does not necessarily coincide with the quantum depolarizing channel

$$\mathcal{D}_p(\rho) = (1-p)\rho + p\frac{I}{2}, \tag{2.99}$$

which represents an isotropic contraction toward the maximally mixed state. Only under additional symmetry or statistical assumptions can the frequency-averaged propagation reduce to such an isotropic model.

The PMD description developed below should therefore be interpreted more generally as frequency-induced polarization decoherence arising from an underlying unitary polarization–frequency evolution [5, 9, 21].

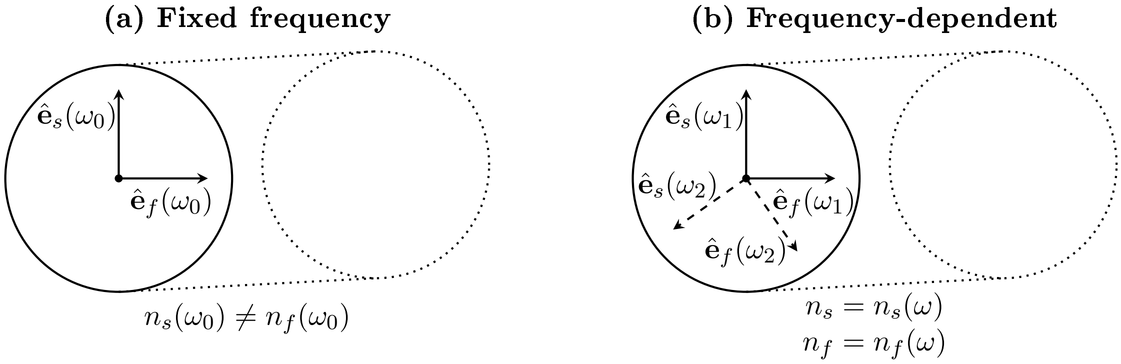

Figure 2.2: Schematic representation of local fiber birefringence at a fixed frequency and over a finite spectral bandwidth. At ω_0, the fiber supports two orthogonal slow and fast polarization eigenmodes characterized by different effective refractive indices. When the frequency dependence of birefringence is taken into account, different spectral components may experience different birefringent parameters; in panel (b), the solid and dashed orthogonal pairs schematically represent the eigenmodes associated with ω_1 and ω_2, respectively.

2.7. Birefringence, Differential Phase, and Group Delay

Polarization-mode dispersion originates from the frequency dependence of birefringence in optical fibers. Even a nominally single-mode fiber supports two orthogonal polarization modes whose propagation constants are generally different [7, 8, 21].

Figure 2.2 schematically illustrates the distinction between a fixed-frequency description of birefringence and the more general frequency-dependent picture. At a given frequency, the fiber is characterized by two orthogonal polarization eigenmodes with different effective refractive indices. Over a finite spectral bandwidth, these birefringent properties may vary with frequency, leading to frequency-dependent polarization evolution.

Consider a locally uniform birefringent segment with slow and fast polarization eigenmodes. Their propagation constants are denoted by $\beta_s(\omega)$ and $\beta_f(\omega)$, respectively. The differential propagation constant is

$$\Delta\beta(\omega) = \beta_s(\omega) - \beta_f(\omega). \tag{2.100}$$

Introducing the corresponding effective refractive indices,

$$\beta_{s,f}(\omega) = \frac{\omega}{c} n_{s,f}(\omega), \tag{2.101}$$

one obtains

$$\Delta\beta(\omega) = \frac{\omega}{c}\Delta n_{\text{eff}}(\omega), \qquad \Delta n_{\text{eff}} = n_s - n_f. \tag{2.102}$$

For a uniform segment of length L_j, the accumulated differential phase is

$$\Delta\phi_j(\omega) = L_j \Delta\beta_j(\omega). \tag{2.103}$$

The corresponding beat length is

$$L_{b,j} = \frac{2\pi}{|\Delta\beta_j|}, \tag{2.104}$$

representing the propagation distance required to accumulate a differential phase of 2π.

Because the differential phase depends on frequency, the two eigenmodes also possess different group delays. The signed group-delay difference of the segment is

$$\Delta\tau_j = \left.\frac{d\Delta\phi_j}{d\omega}\right|_{\omega_0} = L_j \left.\frac{d\Delta\beta_j}{d\omega}\right|_{\omega_0}. \tag{2.105}$$

Using Eq. (2.102),

$$\Delta\tau_j = \frac{L_j}{c}\left[\Delta n_{\text{eff}}(\omega_0) + \omega_0 \left.\frac{d\Delta n_{\text{eff}}}{d\omega}\right|_{\omega_0}\right]. \tag{2.106}$$

The magnitude $|\Delta\tau_j|$ is the differential group delay of the segment. This establishes the physical chain connecting local birefringence to the frequency-dependent polarization evolution:

$$\Delta n_{\text{eff}} \longrightarrow \Delta\beta \longrightarrow \Delta\phi \longrightarrow \frac{d\Delta\phi}{d\omega} \longrightarrow \Delta\tau.$$

2.8. Single Frequency-Dependent Birefringent Segment

The relations introduced in the previous section describe how local birefringence produces a differential phase and, through its frequency dependence, a differential group delay. These quantities can now be incorporated into an operator description of the polarization evolution.

Consider a locally uniform, linear, and lossless birefringent fiber segment of length L_j. Its polarization evolution can be represented by a frequency-dependent unitary Jones operator [22]. With the phase convention adopted in this work,

$$U_j(\omega) = e^{i\bar{\phi}_j(\omega)} \exp\left[\frac{i}{2}\Delta\phi_j(\omega)\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma}\right], \tag{2.107}$$

where $\bar{\phi}_j(\omega)$ is a polarization-independent phase, $\Delta\phi_j(\omega)$ is the differential phase accumulated between the two local polarization eigenmodes, and $\hat{\mathbf{n}}_j$ identifies the local birefringence axis on the Bloch sphere.

The orientation of $\hat{\mathbf{n}}_j$ fixes which of the two orthogonal eigenmodes is associated with the positive eigenvalue of $\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma}$. In the convention used below, the slow mode corresponds to the $+1$ eigenvalue and the fast mode to the -1 eigenvalue. Reversing the orientation $\hat{\mathbf{n}}_j \to -\hat{\mathbf{n}}_j$ changes the sign in the exponential but leaves all physical predictions unchanged provided that the convention is used consistently.

At a fixed frequency, Eq. (2.107) describes a coherent polarization rotation. The connection between this general birefringent evolution and the reduced phase model introduced earlier is made explicit in the following subsection.

2.8.1. Fixed-Frequency Limit and Connection to the Reduced Phase Model

Consider first a monochromatic component at the carrier frequency ω_0. Let $|f_j\rangle$ and $|s_j\rangle$ denote the local fast and slow polarization eigenmodes of the segment, with propagation constants $\beta_{f,j}(\omega_0)$ and $\beta_{s,j}(\omega_0)$, respectively.

An arbitrary input polarization state can be expanded in this eigenmode basis as

$$|\chi_{\text{in}}\rangle = a|f_j\rangle + b|s_j\rangle. \tag{2.108}$$

Because the segment is uniform, the two eigenmodes propagate independently. After a distance L_j,

$$|\chi_{\text{out}}\rangle = ae^{i\beta_{f,j}(\omega_0)L_j}|f_j\rangle + be^{i\beta_{s,j}(\omega_0)L_j}|s_j\rangle. \tag{2.109}$$

Introducing the mean and differential propagation constants,

$$\bar{\beta}_j = \frac{\beta_{s,j} + \beta_{f,j}}{2}, \qquad \Delta\beta_j = \beta_{s,j} - \beta_{f,j}, \tag{2.110}$$

the individual propagation constants can be written as

$$\beta_{f,j} = \bar{\beta}_j - \frac{\Delta\beta_j}{2}, \qquad \beta_{s,j} = \bar{\beta}_j + \frac{\Delta\beta_j}{2}.$$

Using the relation introduced in Eq. (2.103),

$$\Delta\phi_j(\omega_0) = L_j\Delta\beta_j(\omega_0), \tag{2.111}$$

the propagation operator in the local eigenmode basis $\{|f_j\rangle, |s_j\rangle\}$ becomes

$$U_j^{\text{eig}}(\omega_0) = e^{i\bar{\beta}_j L_j}\begin{pmatrix} e^{-i\Delta\phi_j(\omega_0)/2} & 0 \\ 0 & e^{+i\Delta\phi_j(\omega_0)/2} \end{pmatrix}. \tag{2.112}$$

The common factor $e^{i\bar{\beta}_j L_j}$ is a global phase and therefore does not affect polarization observables. The physically relevant transformation is consequently determined by the differential phase $\Delta\phi_j$.

For a generic birefringent segment, the eigenmodes need not coincide with the laboratory horizontal–vertical basis. Let

$$V_j = \big(|f_j\rangle \;\; |s_j\rangle\big) \tag{2.113}$$

denote the unitary matrix whose columns are the two eigenmodes expressed in the laboratory basis. The corresponding fixed-frequency Jones operator is

$$U_j(\omega_0) = e^{i\bar{\beta}_j L_j}V_j\begin{pmatrix} e^{-i\Delta\phi_j/2} & 0 \\ 0 & e^{+i\Delta\phi_j/2} \end{pmatrix}V_j^{\dagger}. \tag{2.114}$$

Equation (2.114) represents the generic fixed-frequency birefringence model. Apart from the polarization-independent phase, it is a unitary rotation around the local birefringence axis $\hat{\mathbf{n}}_j$. It is therefore equivalent to the fixed-frequency limit of Eq. (2.107).

A direct connection with the reduced phase operator can first be established through the special case in which the segment eigenmodes themselves coincide with the computational polarization basis,

$$|f_j\rangle \equiv |H\rangle \equiv |0\rangle, \qquad |s_j\rangle \equiv |V\rangle \equiv |1\rangle. \tag{2.115}$$

In this particular case $V_j = I$, and Eq. (2.112) reduces to

$$U_j(\omega_0) = e^{i\bar{\beta}_j L_j} \begin{pmatrix} e^{-i\Delta\phi_j/2} & 0 \\ 0 & e^{+i\Delta\phi_j/2} \end{pmatrix}. \tag{2.116}$$

Factoring out the common phase associated with the first component gives

$$U_j(\omega_0) = e^{i\left(\bar{\beta}_j L_j - \Delta\phi_j/2\right)} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\Delta\phi_j} \end{pmatrix}. \tag{2.117}$$

Since the prefactor is physically irrelevant for polarization observables, the effective transformation is

$$U_j^{\mathrm{red}}(\theta_j) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta_j} \end{pmatrix}, \qquad \theta_j = \Delta\phi_j(\omega_0) = L_j \Delta\beta_j(\omega_0). \tag{2.118}$$

Equation (2.118) shows algebraically that the reduced phase operator is contained as a special fixed-frequency case of the general birefringent transformation. This aligned single-segment example, however, should not be interpreted as implying that polarization compensation aligns the local birefringence eigenmodes throughout a real fiber.

In a nonuniform fiber, the local fast and slow axes generally vary along the propagation direction. The fixed-frequency transformation of the complete fiber is instead obtained from the ordered product of the transformations of all its segments,

$$U_{\mathrm{fib}}(\omega_0) = U_N(\omega_0) U_{N-1}(\omega_0) \cdots U_1(\omega_0). \tag{2.119}$$

The polarization-compensation procedure employed in the experimental setting acts on this complete input–output transformation rather than on the internal local eigenmodes of the individual fiber segments. Within the reduced model adopted in this work, compensation realigns the relevant output polarization axis with the computational reference basis while leaving a residual rotation around that stabilized axis. The effective compensated transformation can therefore be represented, up to an overall phase, as

$$U_{\mathrm{fib}}^{\mathrm{comp}}(\omega_0) \sim \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \tag{2.120}$$

where θ represents the residual coherent phase not removed by the compensation procedure.

Thus, the reduced phase model is not a different physical propagation mechanism. Rather, it is a one-parameter effective description of the residual fixed-frequency transformation of the compensated fiber. The aligned single-segment case above provides the simplest mathematical realization of the same operator form, without requiring the local birefringence axes of a real segmented fiber to coincide with the laboratory H/V basis.

For the two-arm configuration considered in this work, let the corresponding residual phases be θ_A and θ_B. Starting from

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right),$$

the local transformation gives

$$\left[U_A^{\mathrm{red}}(\theta_A) \otimes U_B^{\mathrm{red}}(\theta_B)\right]|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(e^{i\theta_B}|01\rangle + e^{i\theta_A}|10\rangle\right) \sim \frac{1}{\sqrt{2}}\left(|01\rangle + e^{i(\theta_A-\theta_B)}|10\rangle\right), \tag{2.121}$$

where $\sim$ denotes equality up to a global phase. Defining

$$\theta = \theta_A - \theta_B, \tag{2.122}$$

one recovers the one-parameter reduced phase state used in the preceding modeling levels.

This fixed-frequency transformation remains fully unitary. It can modify polarization correlations and measurement outcomes, but it cannot by itself reduce the purity of the polarization state. Polarization decoherence emerges only when the frequency dependence of the birefringent transformation becomes relevant over the finite spectral bandwidth of the photon.

2.8.2. First-Order Frequency Dependence and Polarization–Frequency Coupling

The fixed-frequency description can now be extended to a finite-bandwidth photon. Over the relevant spectral interval, the local birefringence axis is assumed to be frequency independent to first order,

$$\hat{\mathbf{n}}_j(\omega) \simeq \hat{\mathbf{n}}_j(\omega_0) \equiv \hat{\mathbf{n}}_j,$$

while the accumulated differential phase retains its frequency dependence.

It is convenient to introduce the angular-frequency offset from the carrier,

$$\Omega = \omega - \omega_0, \tag{2.123}$$

so that

$$\omega = \omega_0 + \Omega. \tag{2.124}$$

The carrier frequency therefore corresponds to $\Omega = 0$. This change of variable will also be adopted in the numerical implementation, where the spectral grids are constructed directly in terms of frequency offsets rather than absolute optical frequencies.

Expanding the differential phase around the carrier frequency gives

$$\Delta\phi_j(\omega_0 + \Omega) \simeq \Delta\phi_j(\omega_0) + \Omega\Delta\tau_j, \tag{2.125}$$

where

$$\Delta\tau_j = \left.\frac{d\Delta\phi_j}{d\omega}\right|_{\omega_0} \tag{2.126}$$

is the signed differential group-delay difference introduced in Section 2.7 [22]. Its magnitude $|\Delta\tau_j|$ is the physical differential group delay of the segment.

To isolate the polarization-dependent part of the propagation, consider Eq. (2.107) without the common polarization-independent phase,

$$\widetilde{U}_j(\omega) = \exp\left[\frac{i}{2}\Delta\phi_j(\omega)\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma}\right]. \tag{2.127}$$

Using the frequency-offset variable, the same physical operator can be parameterized as

$$\mathcal{U}_j(\Omega) \equiv \widetilde{U}_j(\omega_0 + \Omega). \tag{2.128}$$

Substituting the first-order expansion of Eq. (2.125) gives

$$\mathcal{U}_j(\Omega) \simeq \exp\left[\frac{i}{2}\left(\Delta\phi_j(\omega_0) + \Omega\Delta\tau_j\right)\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma}\right]. \tag{2.129}$$

Equations (2.127) and (2.129) describe the same transformation: the former uses the absolute optical frequency ω, whereas the latter uses the detuning $\Omega = \omega - \omega_0$.

A compensating transformation that removes the polarization evolution at the carrier frequency is

$$C_j = \widetilde{U}_j^\dagger(\omega_0) = \mathcal{U}_j^\dagger(0). \tag{2.130}$$

Since the birefringence axis is assumed constant over the bandwidth, the carrier transformation and the frequency-dependent transformation commute. The residual operator can therefore be written as

$$\begin{aligned}\mathcal{U}_j^{\mathrm{res}}(\Omega) &\equiv U_j^{\mathrm{res}}(\omega_0 + \Omega)\\ &= C_j\mathcal{U}_j(\Omega)\\ &\simeq \exp\left[\frac{i}{2}\Omega\Delta\tau_j\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma}\right].\end{aligned} \tag{2.131}$$

Thus, once the carrier-frequency transformation has been compensated, the first-order residual polarization evolution depends only on the frequency offset Ω, the signed differential delay $\Delta\tau_j$, and the local birefringence axis $\hat{\mathbf{n}}_j$.

The polarization-independent factor $e^{i\bar{\phi}_j(\omega)}$ has been omitted from Eq. (2.131). For each spectral component it acts identically on both polarization eigenmodes and therefore cancels from the reduced polarization density operator. Its frequency dependence may nevertheless describe polarization-independent temporal propagation and chromatic dispersion, which are outside the polarization-focused model considered here.

Let $|s_+\rangle$ and $|s_-\rangle$ denote the eigenstates of the local birefringence generator,

$$(\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma})\,|s_\pm\rangle = \pm|s_\pm\rangle. \tag{2.132}$$

An input polarization state can be decomposed as

$$|\chi\rangle = \alpha|s_+\rangle + \beta|s_-\rangle. \tag{2.133}$$

Let $f(\omega)$ denote the spectral amplitude of the photon. In terms of the frequency-offset coordinate, define

$$f_\Omega(\Omega) \equiv f(\omega_0 + \Omega). \tag{2.134}$$

Since $d\omega = d\Omega$, the normalization condition becomes

$$\int d\Omega\,|f_\Omega(\Omega)|^2 = 1. \tag{2.135}$$

If polarization and frequency are initially separable, the input wave packet may therefore be written as

$$|\Psi_{\text{in}}\rangle = \int d\Omega\, f_\Omega(\Omega)|\omega_0 + \Omega\rangle \otimes |\chi\rangle. \tag{2.136}$$

Applying the residual operator of Eq. (2.131) gives

$$|\Psi_{\text{out}}\rangle = \int d\Omega\, f_\Omega(\Omega)|\omega_0 + \Omega\rangle \otimes \left[\alpha e^{+i\Omega\Delta\tau_j/2}|s_+\rangle + \beta e^{-i\Omega\Delta\tau_j/2}|s_-\rangle\right]. \tag{2.137}$$

The complete polarization–frequency state remains pure because the propagation is unitary. Unless the input polarization coincides with one of the local birefringence eigenstates, however, the polarization and frequency degrees of freedom are no longer separable.

In the time-domain representation, the opposite frequency-dependent phases in Eq. (2.137) correspond to opposite temporal translations of the two eigenmode components. Their relative temporal separation is the differential group delay $|\Delta\tau_j|$.

This mechanism is illustrated schematically in Fig. 2.3.

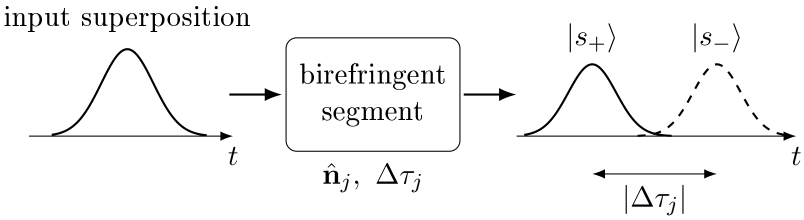

Figure 2.3: Schematic time-domain interpretation of first-order polarization-mode dispersion in a single birefringent segment. An input superposition of the two local polarization eigenmodes evolves into two wave-packet components separated by the differential group delay $|\Delta\tau_j|$. The complete polarization–frequency state remains pure, while reduced polarization coherence decreases when the temporal degree of freedom is not resolved and the overlap between the two components becomes incomplete.

If the spectral degree of freedom is not resolved by the measurement, the polarization state is obtained by tracing it out,

$$\rho_{\text{pol}}^{\text{out}} = \text{Tr}_\omega\left(|\Psi_{\text{out}}\rangle\langle\Psi_{\text{out}}|\right). \tag{2.138}$$

Since the transformation $\omega = \omega_0 + \Omega$ is a simple translation of the spectral coordinate, the same partial trace may equivalently be evaluated over Ω. The diagonal polarization populations remain unchanged, whereas the off-diagonal coherence terms are multiplied by

$$\kappa_j = \int d\Omega\, |f_\Omega(\Omega)|^2 e^{i\Omega\Delta\tau_j}. \tag{2.139}$$

Consequently, in the local eigenmode basis,

$$\rho_{\text{pol}}^{\text{out}} = \begin{pmatrix} \rho_{++}^{\text{in}} & \kappa_j \rho_{+-}^{\text{in}} \\ \kappa_j^* \rho_{-+}^{\text{in}} & \rho_{--}^{\text{in}} \end{pmatrix}. \tag{2.140}$$

The factor κ_j is the characteristic function of the spectral probability density, expressed here in the frequency-offset coordinate and evaluated at the differential delay. Its magnitude therefore quantifies the residual indistinguishability, and hence the polarization coherence, of the two delayed eigenmode components [5].

In the monochromatic limit,

$$|f_\Omega(\Omega)|^2 \to \delta(\Omega),$$

one obtains

$$|\kappa_j| = 1.$$

The polarization state therefore remains fully coherent, consistently with the fixed-frequency unitary model derived in Section 2.8.1.

For a finite bandwidth, $|\kappa_j| < 1$ whenever the spectral averaging makes the two polarization components partially distinguishable in the unresolved temporal degree of freedom. The resulting loss of reduced polarization coherence does not originate from non-unitary evolution of the complete system; it arises from polarization–frequency coupling followed by a partial trace over frequency.

For a Gaussian spectral probability density expressed in terms of the frequency offset,

$$P_\Omega(\Omega) = |f_\Omega(\Omega)|^2 = \frac{1}{\sqrt{2\pi}\sigma_\omega} \exp\left[-\frac{\Omega^2}{2\sigma_\omega^2}\right], \tag{2.141}$$

where σ_ω is the root-mean-square width of the *spectral probability density*, Eq. (2.139) gives

$$|\kappa_j| = \exp\left[-\frac{1}{2}\sigma_\omega^2 \Delta\tau_j^2\right]. \tag{2.142}$$

The dimensionless product

$$\xi_j = \sigma_\omega |\Delta\tau_j|$$

therefore controls the strength of the polarization-coherence reduction. The same differential group delay has little influence on a sufficiently narrowband photon, whereas it can strongly suppress the reduced polarization coherence when the spectral bandwidth becomes sufficiently large [5, 6].

This offset-frequency formulation is also the one adopted in the MATLAB implementation. The numerical routines discretize Ω around zero and evaluate the operator

$\mathcal{U}_j(\Omega) = \widetilde{U}_j(\omega_0 + \Omega)$. The use of Ω therefore represents only a change of spectral coordinate and does not introduce an additional physical approximation beyond the first-order expansion of the differential phase.

This single-segment description provides the elementary physical building block for the concatenated birefringent-fiber model developed in the following section.

2.9. Concatenated Birefringent Fiber Model

The local birefringence of a real optical fiber generally varies along the propagation direction because of geometrical imperfections, residual stress, bending, twisting, and environmental perturbations. A nonuniform fiber can therefore be modeled as a sequence of locally uniform birefringent segments [7, 8, 21].

Figure 2.4 illustrates this representation.

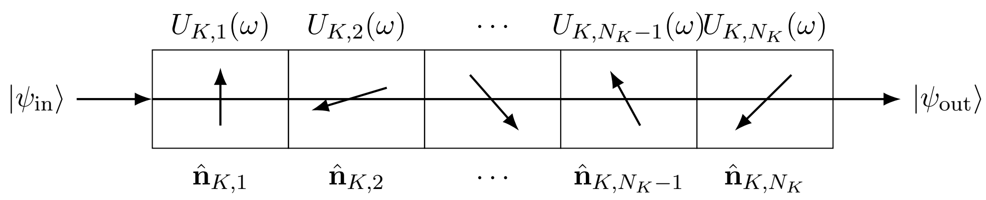

Figure 2.4: Segmented representation of a nonuniform birefringent fiber. Each section is characterized by its own birefringence axis and frequency-dependent unitary operator. The different axis orientations schematically represent the spatial variation of local birefringence along the fiber. The total fiber transformation is the ordered product of the individual segment operators.

For arm K, let N_K denote the number of segments. If the photon encounters segment 1 first and segment N_K last, the complete transformation is

$$U_K(\omega_K) = U_{K,N_K}(\omega_K) \cdots U_{K,2}(\omega_K) U_{K,1}(\omega_K). \tag{2.143}$$

The spatial ordering is essential because transformations associated with different birefringence axes generally do not commute:

$$U_{K,j}(\omega) U_{K,k}(\omega) \neq U_{K,k}(\omega) U_{K,j}(\omega). \tag{2.144}$$

Each segment therefore changes the polarization state on which all subsequent segments act.

If all segments share the same oriented birefringence axis, the operators commute and the differential phases add directly. In this special case, the complete fiber reduces to an equivalent single birefringent element. In the general case, the complete ordered product must be retained.

2.9.1. Fixed and Statistical Fiber Realizations

A particular segmented fiber is defined by the collection of parameters associated with its constituent sections. A realization of arm K may be represented schematically as

$$\mathcal{R}_K = \{\hat{\mathbf{n}}_{K,j}, \Delta\phi_{K,j}(\omega_{0,K}), \Delta\tau_{K,j}\}_{j=1}^{N_K}. \tag{2.145}$$

For a fixed realization, all of these quantities are prescribed and the frequency-dependent operator $U_K(\omega)$ is deterministic.

A statistical fiber model instead samples some or all segment parameters from specified probability distributions. Such sampling represents uncertainty or variation in the spatial birefringence profile. The particular statistical distributions and numerical discretization adopted in the simulator are introduced in the numerical framework chapter.

It is important to distinguish spatial randomness from quantum-state mixedness. For one fixed fiber realization and one resolved frequency pair, propagation through the entire segmented fiber remains unitary. Random orientation of the segments along the fiber does not by itself make the propagated state mixed.

Mixedness in the reduced polarization state can instead arise from:

- tracing over unresolved spectral or temporal degrees of freedom;
- statistical averaging over different fiber realizations if the physical fiber changes during the observation interval.

These two averaging procedures describe different physical mechanisms and are therefore kept conceptually distinct throughout the model.

2.10. Principal States of Polarization and Differential Group Delay

The local birefringence eigenmodes of the individual segments do not generally coincide with the principal states of polarization (PSPs) of the complete fiber. The effective first-order PMD properties must instead be extracted from the frequency dependence of the complete ordered Jones operator [7, 21].

The complete operator is decomposed into a polarization-independent phase and an SU(2) polarization transformation,

$$U_K(\omega) = e^{i\Phi_K(\omega)} W_K(\omega), \qquad W_K(\omega) \in \mathrm{SU}(2). \tag{2.146}$$

The derivative of the scalar phase $\Phi_K(\omega)$ represents a group-delay contribution common to both polarization components. It does not affect the reduced polarization state, although it may influence the absolute arrival time of the optical wave packet.

The polarization-dependent first-order response is described by the input-side PMD generator

$$G_K^{\mathrm{in}}(\omega) = -i W_K^{\dagger}(\omega) \frac{\partial W_K(\omega)}{\partial \omega}. \tag{2.147}$$

Because W_K is unitary with unit determinant, G_K^{in} is Hermitian and traceless. It can therefore be expanded in the Pauli basis as

$$G_K^{\mathrm{in}} = \frac{1}{2} \boldsymbol{\tau}_K \cdot \boldsymbol{\sigma}, \tag{2.148}$$

where $\boldsymbol{\tau}_K$ is the PMD vector.

Its magnitude defines the non-negative first-order differential group delay,

$$\tau_{\mathrm{DGD},K} = \|\boldsymbol{\tau}_K\|. \tag{2.149}$$

Equivalently, the eigenvalues of G_K^{in} are

$$\lambda_{\pm} = \pm \frac{\tau_{\mathrm{DGD},K}}{2}. \tag{2.150}$$

The corresponding eigenvectors,

$$|p^{\mathrm{in}}_{K,+}\rangle, \qquad |p^{\mathrm{in}}_{K,-}\rangle, \tag{2.151}$$

define the two input principal states of polarization. The corresponding output PSPs at the carrier frequency are

$$|p^{\mathrm{out}}_{K,\pm}\rangle = W_K(\omega_{0,K})|p^{\mathrm{in}}_{K,\pm}\rangle. \tag{2.152}$$

To first order around the carrier frequency,

$$W_K(\omega) \simeq W_K(\omega_{0,K}) \exp\left[i(\omega - \omega_{0,K})G^{\mathrm{in}}_K(\omega_{0,K})\right]. \tag{2.153}$$

A principal state therefore propagates according to

$$W_K(\omega)|p^{\mathrm{in}}_{K,\pm}\rangle \simeq e^{\pm i(\omega-\omega_{0,K})\tau_{\mathrm{DGD},K}/2}|p^{\mathrm{out}}_{K,\pm}\rangle. \tag{2.154}$$

Thus, within the first-order PMD approximation, a photon launched in one PSP emerges with a frequency-independent output polarization. The spectral components differ only by a frequency-dependent phase associated with an early or late temporal displacement.

A superposition of the two PSPs can instead become correlated with arrival time, providing the physical mechanism through which PMD reduces polarization coherence after the unresolved temporal or spectral degree of freedom is traced out.

The PSPs of the complete fiber must not be confused with the local birefringence eigenmodes of individual segments. Local eigenmodes diagonalize the Jones operator of one fiber section, whereas the PSPs diagonalize the first-order frequency response of the complete concatenated transformation.

The direction assigned to the PMD vector depends on the phase and Stokes-space conventions adopted in the literature. The physically invariant quantities used here are the PSP axis and the non-negative DGD $\tau_{\mathrm{DGD},K}$.

2.11. PMD Acting on Polarization-Entangled Photon Pairs

For the two-arm system, the polarization transformation associated with a resolved frequency pair is

$$U_{AB}(\omega_A, \omega_B) = U_A(\omega_A) \otimes U_B(\omega_B). \tag{2.155}$$

The observable polarization state after the unresolved spectral degrees of freedom have been traced out is therefore

$$\rho_{AB}^{\text{pol,out}} = \iint d\omega_A \, d\omega_B \, J(\omega_A, \omega_B) \left[U_A(\omega_A) \otimes U_B(\omega_B)\right] \rho_{AB}^{\text{pol,in}} \left[U_A(\omega_A) \otimes U_B(\omega_B)\right]^\dagger . \tag{2.156}$$

For every individual frequency pair, the polarization transformation remains a product of local unitary operators. The effective reduced polarization map can nevertheless be non-unitary because different spectral components undergo different transformations and are subsequently combined through the spectral trace [5–7, 9].

If the joint spectral intensity factorizes,

$$J(\omega_A, \omega_B) = J_A(\omega_A) J_B(\omega_B), \tag{2.157}$$

the reduced polarization transformation factorizes into independently averaged local channels,

$$\mathcal{E}_{AB} = \mathcal{E}_A \otimes \mathcal{E}_B. \tag{2.158}$$

If the photon frequencies are spectrally correlated, however,

$$J(\omega_A, \omega_B) \neq J_A(\omega_A) J_B(\omega_B), \tag{2.159}$$

the local transformations in the two arms are sampled according to a correlated joint distribution.

The physical action remains local for each frequency pair because

$$U_{AB}(\omega_A, \omega_B) = U_A(\omega_A) \otimes U_B(\omega_B).$$

Nevertheless, the effective polarization map does not generally factorize into two independently averaged local channels. The spectral correlations inherited from the SPDC source therefore induce correlations between the effective polarization noise experienced in the two arms.

This distinction does not imply a physical interaction between the remote photons. It reflects instead the fact that the local frequency-dependent transformations are averaged according to a common correlated spectral distribution.

PMD thus provides a physical realization of the mechanism introduced conceptually earlier in the chapter: an underlying lossless and unitary polarization–frequency evolution can appear as polarization decoherence when the spectral and temporal degrees of freedom are not observed.

2.12. Relation Between the Modeling Levels

The models introduced in this chapter are not competing descriptions of the same propagation process. They form the progressive hierarchy summarized in Fig. 2.1, with different physical information retained at each level.

The fixed relative-phase model isolates coherent polarization evolution after partial compensation. Because the transformation is local and unitary, it changes the orientation of measured correlations without degrading purity or entanglement.

The statistical phase model introduces unresolved variability of the coherent transformation. Ensemble averaging suppresses transverse coherence through the complex parameter

$$\mu = \left\langle e^{i\theta} \right\rangle . \tag{2.160}$$

The magnitude $|\mu|$ quantifies coherence loss, while its phase retains the remaining coherent orientation.

The global depolarizing model provides an isotropic analytical baseline. The local two-arm depolarizing model adapts this description to the physical geometry of separate propagation paths through the parameters p_A and p_B. Their combined action on bipartite correlations is summarized by

$$\eta = (1 - p_A)(1 - p_B). \tag{2.161}$$

The composite effective model combines coherent phase evolution, statistical dephasing, and local isotropic depolarization within a common polarization-only framework.

These effective models remain useful because they provide closed-form analytical predictions, make different degradation signatures transparent, and constitute controlled benchmarks for validating the numerical implementation. They should not, however, be interpreted as complete microscopic models of optical-fiber propagation.

The frequency-dependent framework extends the state space by retaining the spectral degrees of freedom. Local birefringence produces a frequency-dependent phase, whose derivative gives a differential group delay. For nonuniform fibers, the complete response is obtained through an ordered concatenation of birefringent sections, and the principal states and total DGD emerge from the frequency derivative of the complete Jones operator.

In this final description, polarization decoherence is not imposed directly through a phenomenological noise probability. For a fixed lossless fiber realization, the enlarged polarization–frequency state evolves unitarily. Effective polarization mixedness emerges from polarization–frequency coupling followed by a partial trace over spectral degrees of freedom that are not resolved experimentally.

The chapter therefore establishes a continuous modeling path from simple analytically controlled channels to a physical PMD description:

coherent polarization evolution

$\downarrow$

statistical phase averaging

$\downarrow$

effective depolarization

$\downarrow$

composite polarization channel

$\downarrow$

frequency-dependent birefringent propagation

This hierarchy provides the theoretical basis for Chapter ??. The analytically tractable models are used first to validate the state-evolution, channel, correlation, and metric routines. The frequency-dependent model is subsequently implemented through spectral discretization and ordered propagation across birefringent fiber segments, allowing polarization degradation to be computed from the physical model rather than prescribed phenomenologically.

3 | Numerical Experiments and Results

The theoretical framework developed in the previous chapters is implemented and investigated numerically in this chapter. The objective is to move from the analytical description of polarization-entangled photon propagation to a computational framework capable of quantifying how physically relevant fiber and source parameters affect the distributed quantum state.

The numerical analysis is organized around a progressive increase in physical complexity. Controlled limiting cases are considered first in order to validate the implementation against analytical predictions. The study is then extended to frequency-dependent birefringent propagation, two-arm polarization-mode dispersion, and statistical ensembles of randomly birefringent fibers. Particular attention is given to the dependence of the output state on quantities that can be related directly to the physical system, including photon bandwidth, differential group delay, fiber length, birefringence statistics, principal-state orientations, and the spectral correlations of the photon pair.

The purpose of the numerical experiments is therefore not limited to verifying the correctness of the implementation. The simulator is used as an analysis tool to identify the physical regimes in which fiber propagation produces negligible, partial, or substantial degradation of polarization entanglement, and to determine how these regimes are reflected in experimentally accessible quantities. The output states are characterized through polarization correlations, reduced density operators, purity, fidelity, concurrence, and Bell-nonlocality indicators. When the fiber model is stochastic, these quantities are treated statistically over an ensemble of fiber realizations rather than solely through their mean values.

A further objective is to establish a direct connection between the numerical model and the experimental investigation developed later in this work. Whenever possible, the numerical experiments are therefore formulated in terms of parameters that can either be controlled or estimated experimentally. This allows the simulator to provide reference predictions for the interpretation of measured single-photon and coincidence statistics and, more generally, to investigate the range of fiber and source conditions compatible with the preservation of useful polarization entanglement.

3.1. Computational Framework and MATLAB Library

All numerical models presented in this chapter are implemented in MATLAB. This choice allows the simulator to be integrated directly into the pre-existing computational codebase used throughout the project, avoiding the introduction of a separate software environment and facilitating maintenance, validation, and future extensions of the framework.

MATLAB is also particularly well suited to the mathematical structure of the problem considered in this work. Quantum states, density operators, Jones operators, Pauli observables, tensor products, and frequency-dependent propagation matrices are represented directly through complex vectors and matrices. As a consequence, the numerical objects used in the implementation closely reproduce both the notation and the algebraic structure of their theoretical counterparts. For example, single-photon polarization operators are represented by 2×2 matrices, bipartite states and operators by vectors and matrices in $\mathbb{C}^2 \otimes \mathbb{C}^2$, and local bipartite transformations are constructed explicitly through tensor products.

The resulting implementation is organized as a modular MATLAB library rather than as a collection of experiment-specific scripts. Reusable routines are provided for state preparation, operator construction, local and frequency-dependent propagation, quantum-channel application, partial traces, correlation evaluation, state metrics, spectral integration, segmented-fiber construction, statistical sampling, and numerical validation. Higher-level numerical experiments then combine these validated components without redefining the underlying mathematical operations.

This architecture provides a direct correspondence between the theoretical model and its computational realization:

$$\begin{gathered}\text{Physical and Mathematical Model}\\ \downarrow\\ \text{Reusable MATLAB Operators and State Routines}\\ \downarrow\\ \text{Frequency-Resolved Fiber Propagation}\\ \downarrow\\ \text{Spectral and Statistical Averaging}\\ \downarrow\\ \text{Density Operator and Observable Evaluation}\\ \downarrow\\ \text{Numerical Experiments and Experimental Predictions.}\end{gathered}$$

Separating the reusable numerical library from the individual experiment workflows has two main advantages. First, the same implementation of each mathematical operation is used throughout the thesis, reducing duplication and making numerical consistency easier to verify. Second, progressively more complex fiber models can be constructed by composition of previously validated components, following the same hierarchical strategy adopted in the theoretical development.

The numerical study presented in the following sections therefore proceeds from controlled validation cases toward increasingly realistic and statistically characterized models of fiber propagation.

3.2. Experiment 1: Single-Segment PMD and the Bandwidth–DGD Regime

The first numerical experiment investigates the elementary mechanism by which polarization-mode dispersion produces effective polarization decoherence for a finite-bandwidth entangled-photon state. The objective is to isolate the competition between spectral bandwidth and differential group delay (DGD) in a configuration for which an analytical solution is available.

A single carrier-compensated birefringent segment is placed in arm A, while arm B is taken to be ideal. This provides a controlled connection between the single-segment model of Section 2.8.2 and the more general segmented-fiber simulations considered later.

3.2.1. Configuration and Analytical Prediction

The input polarization state is

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right), \qquad \rho_{\text{in}} = |\Psi^+\rangle\langle\Psi^+|. \tag{3.1}$$

The segment in arm A has birefringence axis

$$\hat{\mathbf{n}}_A = \hat{\mathbf{z}},$$

and its carrier-frequency differential phase is compensated,

$$\Delta\phi_A(\omega_{0,A}) = 0.$$

Introducing the frequency offset

$$\Omega_A = \omega_A - \omega_{0,A},$$

the residual transformation is therefore

$$U_A^{\text{res}}\left(\omega_{0,A} + \Omega_A\right) = \exp\left[\frac{i}{2}\Omega_A \Delta\tau_A \sigma_z\right], \tag{3.2}$$

with

$$\tau_{\text{DGD}} = |\Delta\tau_A|.$$

Arm B undergoes the identity transformation. The zero carrier phase ensures that the experiment isolates frequency-induced decoherence without introducing an additional coherent rotation that could independently modify the fidelity.

The source is represented by the Gaussian joint spectral intensity

$$J(\Omega_A, \Omega_B) \propto \exp\left[-\frac{(\Omega_A + \Omega_B)^2}{2\sigma_\Sigma^2} - \frac{(\Omega_A - \Omega_B)^2}{2\sigma_\Delta^2}\right]. \tag{3.3}$$

For this experiment,

$$\sigma_\Sigma = \sigma_\Delta = \sqrt{2}\sigma_\omega, \tag{3.4}$$

so that the JSI factorizes and both photons have independent Gaussian marginal spectra with root-mean-square width σ_ω.

For photon A, the spectral trace produces the coherence factor derived in Section 2.8.2,

$$\kappa = \exp\left[-\frac{1}{2}\sigma_\omega^2\tau_{\mathrm{DGD}}^2\right]. \tag{3.5}$$

Introducing the dimensionless PMD strength

$$\xi = \sigma_\omega\tau_{\mathrm{DGD}}, \tag{3.6}$$

the result becomes

$$\boxed{\kappa(\xi) = e^{-\xi^2/2}}. \tag{3.7}$$

The reduced polarization state is consequently

$$\rho_{\mathrm{pol}}^{\mathrm{out}} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & \kappa & 0 \\ 0 & \kappa & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{3.8}$$

with correlation tensor

$$T = \begin{pmatrix} \kappa & 0 & 0 \\ 0 & \kappa & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{3.9}$$

The expected state metrics are therefore

$$\gamma = \frac{1+\kappa^2}{2}, \qquad F_{\Psi^+} = \frac{1+\kappa}{2}, \qquad C = \kappa, \qquad \tau = \kappa^2, \tag{3.10}$$

and

$$S_{\max} = 2\sqrt{1+\kappa^2}. \tag{3.11}$$

Thus, for $\xi = 0$, the original pure maximally entangled state is recovered, whereas for $\xi \to \infty$,

$$C \to 0, \qquad \gamma \to \frac{1}{2}, \qquad F_{\Psi^+} \to \frac{1}{2}, \qquad S_{\max} \to 2.$$

3.2.2. Numerical Procedure

The numerical implementation evaluates the frequency-dependent propagation on a discrete grid of angular-frequency offsets. For every value of σ_ω, the interval

$$\Omega \in [-6\sigma_\omega, +6\sigma_\omega]$$

is sampled using 81 uniformly spaced nodes per arm. The corresponding normalized joint spectral weights Q_{kl} are constructed from Eq. (3.3).

For every spectral pair, the propagated polarization state is

$$\rho_{kl} = \left[U_A(\omega_{0,A} + \Omega_{A,k}) \otimes I_2\right] \rho_{\text{in}} \left[U_A(\omega_{0,A} + \Omega_{A,k}) \otimes I_2\right]^\dagger, \tag{3.12}$$

and the unresolved spectrum is traced out numerically through

$$\rho_{\text{pol}}^{\text{num}} = \sum_{k,l} Q_{kl}\rho_{kl}. \tag{3.13}$$

The parameter range used in the experiment is summarized in Table 3.1.

Table 3.1: Numerical parameters used in Experiment 1.

Parameter	Range or value	Sampling
DGD	0–20 ps	61 points
σ_ω	2.5×10^{10}–2.0×10^{11} rad/s	21 points
$\sigma_f = \sigma_\omega/(2\pi)$	approximately 4–32 GHz	derived
Spectral interval	$\pm 6\sigma_\omega$	81 nodes per arm
$\Delta\phi_A(\omega_{0,A})$	0	fixed
$\hat{\mathbf{n}}_A$	$\hat{\mathbf{z}}$	fixed

The resulting sweep contains 1281 bandwidth–DGD combinations and reaches

$$\xi_{\max} = (2.0 \times 10^{11})(20 \times 10^{-12}) = 4,$$

thereby covering weak, intermediate, and strong polarization-coherence suppression.

3.2.3. Results and Analytical Comparison

Figure 3.1 shows the numerical dependence of concurrence, global purity, Bell-state fidelity, and maximal CHSH parameter on DGD and spectral bandwidth. The bandwidth is expressed as

$$\sigma_f = \frac{\sigma_\omega}{2\pi}$$

in gigahertz.

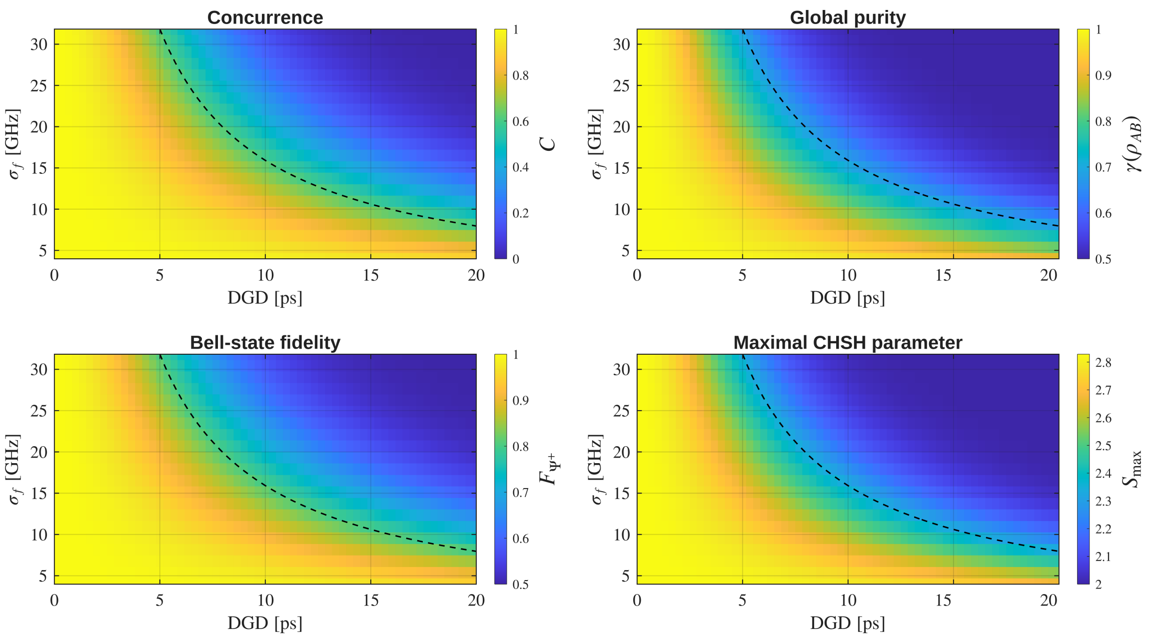

Figure 3.1: Bandwidth–DGD regime maps for a single carrier-compensated birefringent segment acting on arm A. The dashed curve denotes $\xi = \sigma_\omega \tau_{\text{DGD}} = 1$. Increasing either spectral bandwidth or DGD produces the same progressive loss of unresolved polarization coherence through their dimensionless product ξ.

The numerical results reproduce the expected dependence on $\xi = \sigma_\omega \tau_{\text{DGD}}$. In the weak-PMD region $\xi \ll 1$, the polarization transformation varies negligibly over the occupied spectrum and the state remains approximately pure and maximally entangled. As ξ increases, different spectral components acquire increasingly different relative phases. After the spectral trace, the transverse coherence decreases, leading simultaneously to reduced purity, fidelity, and concurrence.

The dashed contour $\xi = 1$ identifies the region in which the DGD becomes comparable with the inverse spectral bandwidth. It is not a sharp threshold, but provides a useful reference between weak and substantial PMD-induced coherence reduction.

In the strong-PMD limit, the state approaches

$$\rho_{\text{pol}}^{\text{out}} \longrightarrow \frac{1}{2}\left(|01\rangle\langle 01| + |10\rangle\langle 10|\right), \tag{3.14}$$

so that the transverse correlations vanish while

$$T_{zz} = -1$$

remains unchanged. The state therefore retains classical anticorrelation in the computational basis even after the polarization entanglement has disappeared.

For this particular state family,

$$S_{\max} = 2\sqrt{1 + C^2},$$

and therefore $S_{\max} > 2$ for every finite value of ξ, approaching the classical limit only asymptotically as $C \to 0$. This behavior is a property of the ideal single-axis dephasing model and should not be generalized to more complex fiber channels.

The point-by-point numerical values were also compared with the analytical predictions. The maximum absolute discrepancies over the complete sweep are reported in Table 3.2.

The agreement at approximately 10^{-9} or better confirms the numerical implementation of the frequency-dependent segment operator, spectral discretization, spectral partial trace, and subsequent state characterization. The remaining discrepancy is consistent with the finite spectral interval and frequency-grid discretization.

3.2.4. Physical Interpretation

The experiment shows that DGD alone is not sufficient to determine the influence of PMD on an entangled photon. The same differential delay may produce negligible degradation

Table 3.2: Maximum absolute numerical–analytical discrepancies in Experiment 1.

Quantity	Maximum absolute error
Purity	3.512×10^{-9}
Fidelity	1.978×10^{-9}
Concurrence	3.956×10^{-9}
Tangle	7.025×10^{-9}
Entanglement of formation	5.471×10^{-9}
$S_{\max}$	5.246×10^{-9}
T_{xx}	3.956×10^{-9}
T_{yy}	3.956×10^{-9}
T_{zz}	1.332×10^{-15}

for a narrow spectrum and substantial degradation for a broader one. The relevant scale is instead

$$\boxed{\xi = \sigma_\omega \tau_{\mathrm{DGD}}}.$$

This result also establishes a direct connection with the effective dephasing model introduced in Section 2.2. In the present physical PMD model, the effective coherence parameter is not assigned phenomenologically but follows from measurable source and propagation parameters,

$$|\mu| \longleftrightarrow |\kappa| = \exp\left[-\frac{1}{2}\sigma_\omega^2 \tau_{\mathrm{DGD}}^2\right]. \tag{3.15}$$

Experiment 1 therefore provides the first quantitative connection between the polarization-only effective models and frequency-dependent fiber propagation. Its assumptions are intentionally restrictive: PMD acts in only one arm, the birefringence axis is fixed, and the photon frequencies are uncorrelated. These restrictions are relaxed in the following experiment, where PMD acts in both arms and the role of the joint spectral correlations is investigated.

3.3. Experiment 2: Two-Arm PMD and Spectral Correlations

3.3.1. Objective and Physical Motivation

Experiment 1 isolated the elementary decoherence mechanism produced by PMD in a single arm. The present experiment extends the analysis to the case in which both photons propagate through frequency-dependent birefringent elements and investigates the additional role of correlations in the two-photon spectrum.

The main objective is to determine whether the spectral correlation between the frequency offsets of the two photons can modify the degradation of polarization entanglement produced by two-arm PMD.

The reference polarization state is again

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right). \tag{3.16}$$

Both arms are assigned the same differential group delay,

$$\tau_A = \tau_B = \tau, \tag{3.17}$$

while their birefringence axes are chosen to be opposite,

$$\hat{\mathbf{n}}_A = +\hat{\mathbf{z}}, \qquad \hat{\mathbf{n}}_B = -\hat{\mathbf{z}}. \tag{3.18}$$

The carrier-frequency differential phases are set to zero, so that the experiment isolates the frequency-dependent contribution responsible for PMD.

3.3.2. Spectral Model and Analytical Prediction

The two photons are assigned equal marginal angular-frequency standard deviations

$$\sigma_A = \sigma_B = \sigma_\omega, \tag{3.19}$$

while their joint spectrum is varied through the spectral correlation coefficient

$$r = \frac{\mathrm{Cov}(\Omega_A, \Omega_B)}{\sigma_\omega^2}. \tag{3.20}$$

The Gaussian widths along the sum and difference frequency coordinates are therefore

$$\sigma^2_{\text{sum}} = 2\sigma^2_\omega(1+r), \tag{3.21}$$

$$\sigma^2_{\text{difference}} = 2\sigma^2_\omega(1-r). \tag{3.22}$$

Negative values of r correspond to frequency anticorrelation, $r = 0$ to an uncorrelated joint spectrum, and positive values to positively correlated frequency offsets.

For the configuration in Eq. (3.18), the frequency-dependent relative phase accumulated between the two components of the Bell state is

$$\theta(\Omega_A, \Omega_B) = \tau(\Omega_A + \Omega_B). \tag{3.23}$$

Its variance is consequently

$$\begin{aligned} \mathrm{Var}(\theta) &= \tau^2\, \mathrm{Var}(\Omega_A + \Omega_B) \\ &= 2\sigma^2_\omega \tau^2 (1+r). \end{aligned} \tag{3.24}$$

For a zero-mean Gaussian spectrum, spectral averaging therefore gives the residual polarization coherence

$$g(\tau, r) = \left\langle e^{i\theta} \right\rangle = \exp\left[-\sigma^2_\omega \tau^2 (1+r)\right]. \tag{3.25}$$

Introducing the normalized PMD strength

$$\xi = \sigma_\omega \tau, \tag{3.26}$$

the result becomes

$$\boxed{g(\xi, r) = e^{-(1+r)\xi^2}}. \tag{3.27}$$

After tracing over the unresolved spectral degrees of freedom, the polarization state takes the form

$$\rho_{AB}(\xi, r) = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & g & 0 \\ 0 & g & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{3.28}$$

The principal analytical quantities follow directly:

$$C = g, \tag{3.29}$$

$$F_{\Psi^+} = \frac{1+g}{2}, \tag{3.30}$$

$$\gamma(\rho_{AB}) = \frac{1+g^2}{2}, \tag{3.31}$$

$$\gamma(\rho_A) = \gamma(\rho_B) = \frac{1}{2}, \tag{3.32}$$

$$S_{\max} = 2\sqrt{1+g^2}. \tag{3.33}$$

The correlation tensor is

$$T(\xi, r) = \begin{pmatrix} g & 0 & 0 \\ 0 & g & 0 \\ 0 & 0 & -1 \end{pmatrix}, \tag{3.34}$$

showing that PMD suppresses the transverse polarization coherence while the longitudinal anticorrelation remains unchanged.

3.3.3. Numerical Configuration

The numerical experiment evaluates the frequency-resolved two-photon state on a uniform joint spectral grid and subsequently performs the discrete spectral trace. The parameters used in the simulation are summarized in Table 3.3.

For each value of τ, the complete set of frequency-resolved polarization states is computed only once. The different spectral-correlation conditions are then introduced by averaging these states with the corresponding joint spectral weights. This separates the physical fiber transformation from the statistical structure of the two-photon spectrum and avoids redundant propagation calculations.

Table 3.3: Numerical configuration of Experiment 2.

Parameter	Value
Marginal bandwidth σ_ω	1.0×10^{11} rad/s
DGD points	61
Normalized PMD interval	$0 \leq \xi \leq 4$
Physical DGD interval	$0 \leq \tau \leq 40\,\mathrm{ps}$
Spectral-correlation interval	$-0.9 \leq r \leq 0.9$
Spectral-correlation points	37
Frequency nodes per arm	101
Spectral window	$\pm 6\sigma_\omega$
Minimum-width resolution	3.73 grid intervals per σ

3.3.4. State-Metric Results

Figure 3.2 shows the evolution of concurrence, Bell-state fidelity, global polarization purity, and maximal CHSH parameter for representative values of r. Continuous curves denote the analytical predictions, while numerical results are shown by discrete markers.

For $r = 0$, Eq. (3.27) reduces to

$$g(\xi, 0) = e^{-\xi^2}. \tag{3.35}$$

Compared with the single-arm result $e^{-\xi^2/2}$, the coherence therefore decreases more rapidly because both independent PMD contributions enter the relative phase variance.

The situation changes substantially when the frequencies are anticorrelated. For example, at $r = -0.9$,

$$C(\xi, -0.9) = e^{-0.1\xi^2}, \tag{3.36}$$

so that even at $\xi = 4$,

$$C(4, -0.9) = e^{-1.6} \simeq 0.202. \tag{3.37}$$

By contrast, for $r = +0.9$,

$$C(\xi, +0.9) = e^{-1.9\xi^2}, \tag{3.38}$$

and the entanglement rapidly approaches zero.

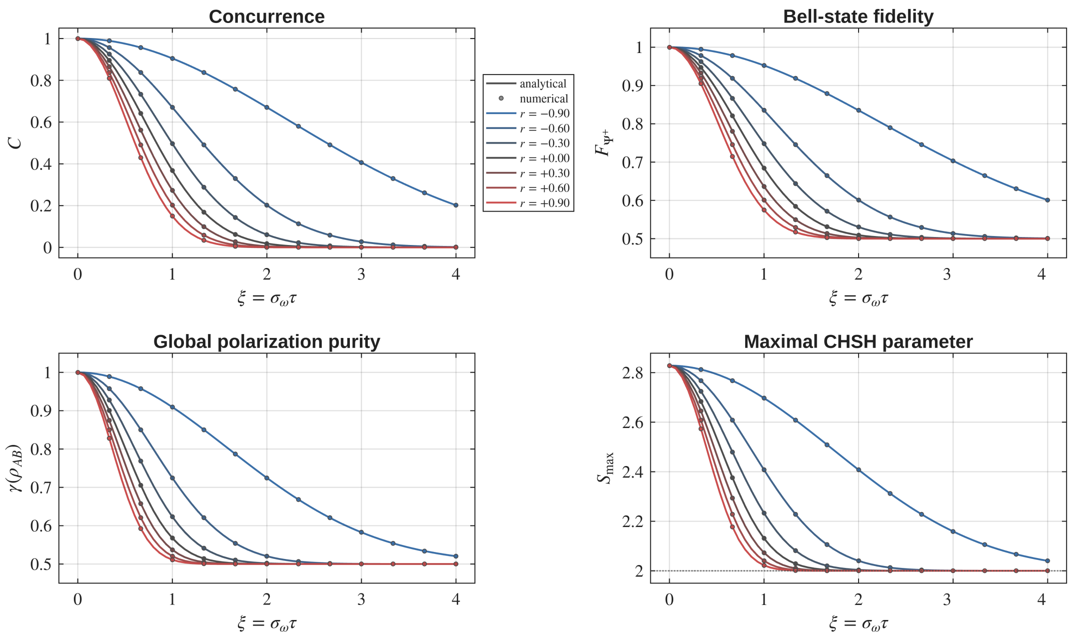

Figure 3.2: State metrics for two-arm PMD as a function of the normalized PMD strength $\xi = \sigma_\omega \tau$, for representative spectral-correlation coefficients r. Analytical predictions are shown as continuous curves and numerical spectral-trace results as markers. Frequency anticorrelation significantly slows the degradation of polarization entanglement, whereas positive spectral correlation accelerates it.

The same behavior is inherited by fidelity, purity, and CHSH nonlocality through Eq. (3.33). In particular, as $g \to 0$,

$$F_{\Psi^+} \to \frac{1}{2}, \qquad \gamma(\rho_{AB}) \to \frac{1}{2}, \qquad S_{\max} \to 2. \tag{3.39}$$

The local states remain maximally mixed throughout the entire experiment, $\rho_A = \rho_B = I/2$. The degradation therefore appears exclusively in the two-photon coherence and correlations rather than through the development of a preferred local polarization state.

3.3.5. Spectral-Correlation Regimes

The combined dependence on PMD strength and spectral correlation is shown in Fig. 3.3.

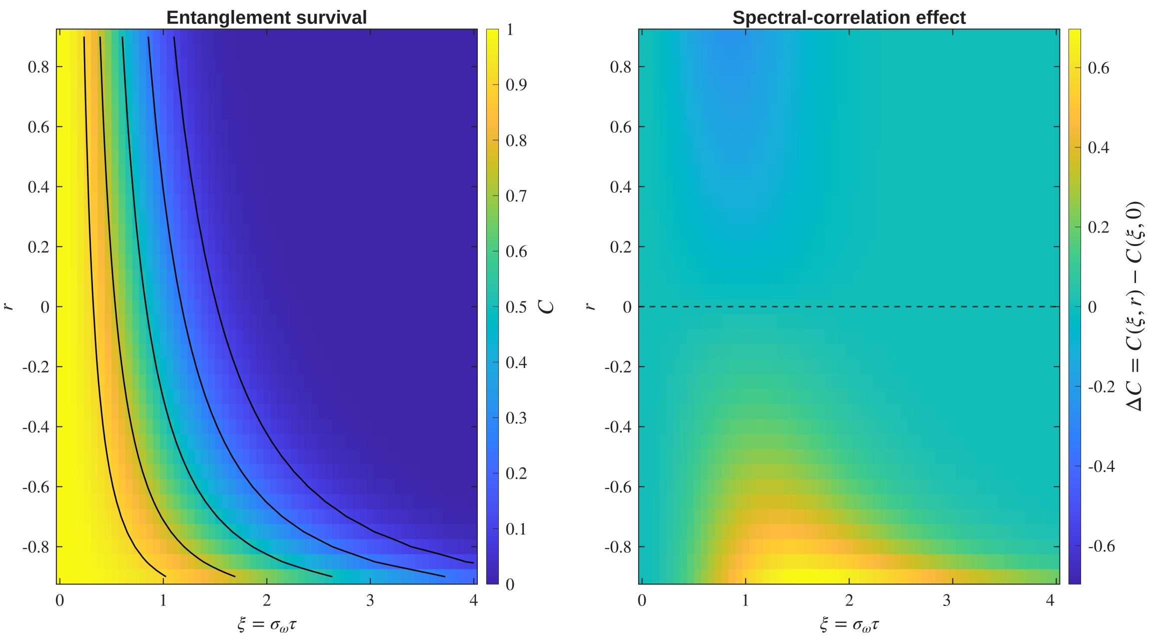

Figure 3.3: Two-arm PMD spectral-correlation regimes. The left panel shows the numerically evaluated concurrence $C(\xi, r)$, with analytical constant-concurrence contours superimposed. The right panel shows the change in concurrence relative to an uncorrelated spectrum, $\Delta C = C(\xi, r) - C(\xi, 0)$. Negative spectral correlation produces a positive entanglement-survival contribution, while positive correlation accelerates PMD-induced degradation.

The regime map makes the role of the joint spectrum particularly clear. Since

$$\mathrm{Var}(\Omega_A + \Omega_B) = 2\sigma_\omega^2(1 + r), \tag{3.40}$$

frequency anticorrelation reduces the variance of the spectral combination to which the polarization state is sensitive. Consequently, the random relative phase is partially suppressed and a larger fraction of the polarization coherence survives the spectral trace.

The effect can be isolated by defining

$$\Delta C(\xi, r) = C(\xi, r) - C(\xi, 0), \tag{3.41}$$

which analytically gives

$$\Delta C(\xi, r) = e^{-(1+r)\xi^2} - e^{-\xi^2}. \tag{3.42}$$

Therefore,

$$\Delta C \begin{cases} > 0, & r < 0, \\ = 0, & r = 0, \\ < 0, & r > 0. \end{cases} \tag{3.43}$$

The improvement for $r < 0$ is not the consequence of a local reduction of the PMD in either fiber. Each photon still undergoes a frequency-dependent polarization transformation. Instead, the joint spectral structure correlates the two local transformations such that their contributions to the relative Bell-state phase partially cancel.

In the ideal limit $r \to -1$,

$$\Omega_A + \Omega_B \to 0, \tag{3.44}$$

and therefore

$$g(\xi, r) \to 1. \tag{3.45}$$

The polarization entanglement can then remain unaffected despite finite PMD in both arms. This constitutes a nonlocal compensation effect originating from spectral correlations rather than from an active polarization correction applied after propagation.

3.3.6. Numerical Validation

The numerical spectral trace was compared over the complete (ξ, r) parameter domain with the continuous Gaussian expressions derived above. The maximum absolute discrepancies are reported in Table 3.4.

Table 3.4: Maximum absolute numerical errors in Experiment 2 relative to the continuous Gaussian analytical solution.

Quantity	Maximum absolute error
Global purity	6.584×10^{-9}
Local purity, arm A	9.326×10^{-15}
Local purity, arm B	9.326×10^{-15}
Bell-state fidelity	3.728×10^{-9}
Concurrence	7.456×10^{-9}
Tangle	1.317×10^{-8}
Entanglement of formation	1.031×10^{-8}
Maximum CHSH parameter	9.870×10^{-9}
c_{xx}	7.456×10^{-9}
c_{yy}	7.456×10^{-9}
c_{zz}	9.326×10^{-15}
Correlation tensor, Frobenius norm	1.054×10^{-8}

All discrepancies remain of order 10^{-8} or smaller, while the longitudinal correlation and local purities agree to approximately machine precision. The agreement confirms that the finite spectral grid and discrete joint-spectrum averaging accurately reproduce the continuous Gaussian model over the complete parameter range considered.

3.3.7. Physical Interpretation

Experiment 2 demonstrates that PMD-induced polarization decoherence cannot in general be characterized by the marginal bandwidths and individual fiber DGDs alone. When both photons propagate through frequency-dependent channels, the joint spectral statistics determine how the two local transformations combine after the spectral degrees of freedom are discarded.

For the opposite-axis configuration considered here, the relevant spectral variable is the sum $\Omega_A + \Omega_B$. Anticorrelated photon pairs exhibit reduced fluctuations of this quantity and therefore retain a larger polarization coherence. Positively correlated photons instead increase the corresponding phase variance and accelerate entanglement degradation.

The experiment thus provides a direct numerical illustration of an important difference between independent and correlated effective channels. Even though the fiber transformations remain local,

$$U_A(\Omega_A) \otimes U_B(\Omega_B), \tag{3.46}$$

their effective action on the reduced polarization state is not independent when Ω_A and Ω_B are statistically correlated.

Spectral correlations can therefore act as a resource for entanglement distribution: they do not eliminate PMD locally, but can strongly suppress its observable effect on the distributed bipartite polarization state.

Bibliography

[1] Stephanie Wehner, David Elkouss, and Ronald Hanson. Quantum internet: A vision for the road ahead. *Science*, 362(6412):eaam9288, 2018. doi: 10.1126/science.aam9288.

[2] Charles H. Bennett and Gilles Brassard. Quantum cryptography: Public key distribution and coin tossing. In *Proceedings of the IEEE International Conference on Computers, Systems and Signal Processing*, pages 175–179, 1984.

[3] Charles H. Bennett, Gilles Brassard, and N. David Mermin. Quantum cryptography without Bell's theorem. *Physical Review Letters*, 68(5):557–559, 1992. doi: 10.1103/PhysRevLett.68.557.

[4] Evan Dowling, Mark Morris, Gerald Baumgartner, Rajarshi Roy, and Thomas E. Murphy. Non-local polarization alignment and control in fibers using feedback from correlated measurements of entangled photons. *Optics Express*, 31(2):2316–2329, 2023. doi: 10.1364/OE.475465.

[5] Phoenix S. Y. Poon and C. K. Law. Polarization and frequency disentanglement of photons via stochastic polarization mode dispersion. *Physical Review A*, 77(3):032327, 2008. doi: 10.1103/PhysRevA.77.032327.

[6] Misha Brodsky, Elizabeth C. George, Cristian Antonelli, and Mark Shtaif. Loss of polarization entanglement in a fiber-optic system with polarization mode dispersion in one optical path. *Optics Express*, 19(1):45–54, 2011. doi: 10.1364/OE.19.000045.

[7] Gabriele Riccardi. *Polarization-Entanglement Distribution in Fiber-Optic Channels*. PhD thesis, Università degli Studi dell'Aquila, L'Aquila, Italy, 2021. Doctoral Program in Physical and Chemical Sciences, XXXIII cycle, academic year 2019/2020.

[8] P. K. A. Wai and C. R. Menyuk. Polarization mode dispersion, decorrelation, and diffusion in optical fibers with randomly varying birefringence. *Journal of Lightwave Technology*, 14(2):148–157, 1996. doi: 10.1109/50.482256.

[9] Cristian Antonelli, Mark Shtaif, and Misha Brodsky. Sudden death of entanglement

induced by polarization mode dispersion. *Physical Review Letters*, 106(8):080404, 2011. doi: 10.1103/PhysRevLett.106.080404.

[10] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, Cambridge, 10th anniversary edition edition, 2010. ISBN 9781107002173.

[11] Karl Kraus. General state changes in quantum theory. *Annals of Physics*, 64(2): 311–335, 1971. doi: 10.1016/0003-4916(71)90108-4.

[12] Joseph B. Altepeter, Daniel F. V. James, and Paul G. Kwiat. Quantum state tomography. In Matteo G. A. Paris and Jaroslav Řeháček, editors, *Quantum State Estimation*, pages 113–145. Springer, Berlin, Heidelberg, 2004. doi: 10.1007/978-3-540-44481-7_4.

[13] William K. Wootters. Entanglement of formation of an arbitrary state of two qubits. *Physical Review Letters*, 80(10):2245–2248, 1998. doi: 10.1103/PhysRevLett.80.2245.

[14] John S. Bell. On the Einstein Podolsky Rosen paradox. *Physics*, 1(3):195–200, 1964.

[15] John F. Clauser, Michael A. Horne, Abner Shimony, and Richard A. Holt. Proposed experiment to test local hidden-variable theories. *Physical Review Letters*, 23(15): 880–884, 1969. doi: 10.1103/PhysRevLett.23.880.

[16] Ryszard Horodecki, Paweł Horodecki, and Michał Horodecki. Violating Bell inequality by mixed spin-1/2 states: Necessary and sufficient condition. *Physics Letters A*, 200(5):340–344, 1995. doi: 10.1016/0375-9601(95)00214-N.

[17] Warren P. Grice and Ian A. Walmsley. Spectral information and distinguishability in type-II down-conversion with a broadband pump. *Physical Review A*, 56(2):1627–1634, 1997. doi: 10.1103/PhysRevA.56.1627.

[18] Peter J. Mosley, Jeff S. Lundeen, Brian J. Smith, Piotr Wasylczyk, Alfred B. U'Ren, Christine Silberhorn, and Ian A. Walmsley. Heralded generation of ultrafast single photons in pure quantum states. *Physical Review Letters*, 100(13):133601, 2008. doi: 10.1103/PhysRevLett.100.133601.

[19] Travis S. Humble and Warren P. Grice. Effects of spectral entanglement in polarization-entanglement swapping and type-i fusion gates. *Physical Review A*, 77 (2):022312, 2008. doi: 10.1103/PhysRevA.77.022312.

[20] Hou Shun Poh, Chune Yang Lum, Ivan Marcikic, Antía Lamas-Linares, and Christian Kurtsiefer. Joint spectrum mapping of polarization entanglement in spontaneous

parametric down-conversion. *Physical Review A*, 75(4):043816, 2007. doi: 10.1103/PhysRevA.75.043816.

[21] J. P. Gordon and H. Kogelnik. PMD fundamentals: Polarization mode dispersion in optical fibers. *Proceedings of the National Academy of Sciences of the United States of America*, 97(9):4541–4550, 2000. doi: 10.1073/pnas.97.9.4541.

[22] Denis Penninckx and Vincent Morenas. Jones matrix of polarization mode dispersion. *Optics Letters*, 24(13):875–877, 1999. doi: 10.1364/OL.24.000875.

A | Foundations of Bipartite Quantum States

A.1. Explicit Derivation of Reduced Density Operators and Comparison of Global States

This appendix provides an explicit derivation of the reduced density operators associated with the phase-evolved Bell state introduced in Chapter 3. It also illustrates an important conceptual point: identical local reduced states may correspond to physically different global states.

The discussion clarifies why local measurements alone are insufficient to characterize bipartite quantum systems and motivates the introduction of global quantities such as purity, correlation functions, and entanglement measures.

Phase-Evolved Bell State

We consider the two-qubit state

$$|\psi(\theta)\rangle = \frac{|01\rangle + e^{i\theta}|10\rangle}{\sqrt{2}}, \tag{A.1}$$

written in the computational basis

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}.$$

Its vector representation is

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ e^{i\theta} \\ 0 \end{pmatrix}. \tag{A.2}$$

The corresponding density operator is

$$\rho_{AB}(\theta) = |\psi(\theta)\rangle\langle\psi(\theta)|. \tag{A.3}$$

Expanding explicitly gives

$$\rho_{AB}(\theta) = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & e^{-i\theta} & 0 \\ 0 & e^{i\theta} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{A.4}$$

Explicit Evaluation of the Partial Trace

The reduced density operator of subsystem A is obtained through the partial trace over subsystem B:

$$\rho_A(\theta) = \mathrm{Tr}_B\left(\rho_{AB}(\theta)\right). \tag{A.5}$$

Using the computational basis of subsystem B,

$$\{|0\rangle, |1\rangle\},$$

the partial trace can be written explicitly as

$$\rho_A(\theta) = \sum_{b=0}^{1} \left(I_A \otimes \langle b|\right) \rho_{AB}(\theta) \left(I_A \otimes |b\rangle\right). \tag{A.6}$$

We now evaluate the individual contributions.

The diagonal terms give

$$\begin{aligned}
\mathrm{Tr}_B\left(|01\rangle\langle 01|\right) &= |0\rangle\langle 0|, \\
\mathrm{Tr}_B\left(|10\rangle\langle 10|\right) &= |1\rangle\langle 1|.
\end{aligned} \tag{A.7}$$

The off-diagonal terms vanish:

$$\begin{aligned}
\mathrm{Tr}_B\left(|01\rangle\langle 10|\right) &= 0, \\
\mathrm{Tr}_B\left(|10\rangle\langle 01|\right) &= 0,
\end{aligned} \tag{A.8}$$

due to the orthogonality relation

$$\langle 0|1\rangle = 0.$$

Summing all contributions yields

$$\rho_A(\theta) = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I}{2}. \tag{A.9}$$

By symmetry,

$$\rho_B(\theta) = \frac{I}{2}. \tag{A.10}$$

Thus, although the global state depends explicitly on the phase θ, the reduced states do not. The phase appears only in the off-diagonal coherence terms of the global density operator, which are eliminated by the partial trace.

In particular, for $\theta = 0$,

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \tag{A.11}$$

one obtains the same reduced states,

$$\rho_A = \rho_B = \frac{I}{2}. \tag{A.12}$$

Local Equivalence and Global Distinction

The previous result illustrates an important conceptual property of bipartite quantum systems.

Consider the classical mixture

$$\rho^{(\text{class})} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I}{2}. \tag{A.13}$$

At the local level,

$$\rho_A^{(\text{ent})} = \rho^{(\text{class})}, \tag{A.14}$$

and therefore, for any observable O_A,

$$\mathrm{Tr}\left(\rho^{(\text{class})} O_A\right) = \mathrm{Tr}\left(\rho_A^{(\text{ent})} O_A\right). \tag{A.15}$$

Hence, local measurements cannot distinguish classical mixtures from reductions of entangled states.

The distinction emerges only at the global level.

Consider the separable mixed state

$$\rho_{AB}^{(\text{sep})} = \frac{1}{2}|01\rangle\langle 01| + \frac{1}{2}|10\rangle\langle 10|. \tag{A.16}$$

Its reduced states are identical to those of the entangled Bell state:

$$\begin{aligned} \mathrm{Tr}_B\left(\rho_{AB}^{(\text{sep})}\right) &= \frac{I}{2}, \\ \mathrm{Tr}_A\left(\rho_{AB}^{(\text{sep})}\right) &= \frac{I}{2}. \end{aligned} \tag{A.17}$$

However, the separable state lacks the coherence terms present in the entangled density operator.

This difference becomes visible through global quantities.

For the entangled state,

$$\gamma\left(\rho_{AB}(\theta)\right) = 1, \tag{A.18}$$

whereas for the separable mixture,

$$\gamma\left(\rho_{AB}^{(\mathrm{sep})}\right) = \frac{1}{2}. \tag{A.19}$$

A second distinction appears in bipartite correlations.

For the entangled phase-evolved state,

$$\begin{aligned} \langle \sigma_x \otimes \sigma_x \rangle &= \cos\theta, \\ \langle \sigma_y \otimes \sigma_y \rangle &= \cos\theta, \end{aligned} \tag{A.20}$$

whereas for the separable mixture,

$$\begin{aligned} \langle \sigma_x \otimes \sigma_x \rangle &= 0, \\ \langle \sigma_y \otimes \sigma_y \rangle &= 0. \end{aligned} \tag{A.21}$$

Thus, states with identical reduced density operators may differ fundamentally at the global level.

Reduced states encode all locally accessible information, but do not capture global coherence or bipartite entanglement.

A.2. Derivation of Fidelity for a Pure Reference State

This appendix derives the simplified expression for quantum fidelity when one of the two states is pure. The result is used throughout Chapter 4 in the evaluation of fidelity with respect to Bell states.

Beyond its algebraic simplification, this derivation also clarifies the operational interpretation of fidelity as a projection probability onto a target quantum state.

General Definition of Fidelity

Let ρ and σ be density operators acting on a finite-dimensional Hilbert space.

The general definition of quantum fidelity is

$$F(\rho, \sigma) = \left(\mathrm{Tr} \sqrt{\sqrt{\rho}\, \sigma \, \sqrt{\rho}} \right)^2 . \tag{A.22}$$

In the present derivation, we consider the special case in which the reference state is pure:

$$\sigma = |\psi\rangle\langle\psi|, \tag{A.23}$$

with

$$\langle\psi|\psi\rangle = 1. \tag{A.24}$$

By construction, the operator σ is:

- Hermitian,
$$\sigma^\dagger = \sigma,$$
- positive semidefinite,
$$\langle\chi|\sigma|\chi\rangle \geq 0, \qquad \forall\, |\chi\rangle,$$
- normalized,
$$\mathrm{Tr}(\sigma) = 1.$$

Moreover, σ is a rank-one projector onto the one-dimensional subspace spanned by $|\psi\rangle$.

Reduction for a Pure Reference State

Substituting Eq. (A.23) into Eq. (A.22) gives

$$\sqrt{\rho}\, \sigma \, \sqrt{\rho} = \sqrt{\rho}\, |\psi\rangle\langle\psi|\, \sqrt{\rho}. \tag{A.25}$$

Define the vector

$$|\phi\rangle = \sqrt{\rho}\, |\psi\rangle. \tag{A.26}$$

Then Eq. (A.25) becomes

$$\sqrt{\rho}\, \sigma \, \sqrt{\rho} = |\phi\rangle\langle\phi|. \tag{A.27}$$

The operator $|\phi\rangle\langle\phi|$ is again rank one and positive semidefinite.

Spectral Structure of the Rank-One Operator

The only nonzero eigenvalue of the operator in Eq. (A.27) is

$$\lambda = \langle\phi|\phi\rangle. \tag{A.28}$$

Introducing the normalized vector

$$|\hat{\phi}\rangle = \frac{|\phi\rangle}{\|\phi\|}, \tag{A.29}$$

the spectral decomposition becomes

$$|\phi\rangle\langle\phi| = \lambda\, |\hat{\phi}\rangle\langle\hat{\phi}|. \tag{A.30}$$

The square root of the operator is therefore

$$\sqrt{|\phi\rangle\langle\phi|} = \sqrt{\lambda}\, |\hat{\phi}\rangle\langle\hat{\phi}|. \tag{A.31}$$

Taking the trace gives

$$\mathrm{Tr}\, \sqrt{|\phi\rangle\langle\phi|} = \sqrt{\lambda}. \tag{A.32}$$

Substituting into Eq. (A.22), one obtains

$$F(\rho,\sigma) = \lambda = \langle\phi|\phi\rangle. \tag{A.33}$$

Using Eq. (A.26),

$$\lambda = \langle\psi|\rho|\psi\rangle. \tag{A.34}$$

Final Expression and Interpretation

Combining the previous results, the fidelity simplifies to

$$F(\rho, |\psi\rangle\langle\psi|) = \langle\psi|\rho|\psi\rangle. \tag{A.35}$$

This is the expression used throughout Chapter 4 for fidelity with respect to Bell states.

The result admits a direct operational interpretation. Since

$$|\psi\rangle\langle\psi|$$

is the projector onto the state $|\psi\rangle$, the fidelity corresponds to the probability of obtaining the outcome $|\psi\rangle$ in a projective measurement performed on the state ρ.

Fidelity therefore quantifies the overlap between the physical state produced by the system and the chosen target reference state.

A.3. Derivations for Concurrence and Reduced Density Operators

This appendix derives the relations connecting concurrence, reduced density operators, and subsystem purity for pure bipartite two-qubit states.

The derivation clarifies how bipartite entanglement is encoded in the mixedness of the reduced subsystems, providing the theoretical foundation for the relations used throughout Chapter 4.

Coefficient-Matrix Representation

Consider a normalized two-qubit pure state written in the computational basis as

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle. \tag{A.36}$$

Introduce the coefficient matrix

$$A_\psi = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \tag{A.37}$$

so that the state can be expressed compactly as

$$|\psi\rangle = \sum_{i,j=0}^{1} (A_\psi)_{ij} \, |i\rangle \otimes |j\rangle. \tag{A.38}$$

This matrix representation provides a convenient bridge between the bipartite wavefunction and the reduced density operators.

Reduced Density Operators

The global density operator is

$$\rho_{AB} = |\psi\rangle\langle\psi|. \tag{A.39}$$

Expanding Eq. (A.38) yields

$$\rho_{AB} = \sum_{i,j,k,l} (A_\psi)_{ij} (A_\psi)^*_{kl} \, |i\rangle\langle k| \otimes |j\rangle\langle l|. \tag{A.40}$$

The reduced density operator on subsystem A is defined by the partial trace

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}). \tag{A.41}$$

Applying the partial trace to Eq. (A.40) gives

$$\rho_A = \sum_{i,j,k,l} (A_\psi)_{ij} (A_\psi)^*_{kl} \left(\sum_v \langle v|j\rangle\langle l|v\rangle \right) |i\rangle\langle k|. \tag{A.42}$$

Using the orthogonality relation

$$\sum_v \langle v|j\rangle\langle l|v\rangle = \delta_{jl}, \tag{A.43}$$

one obtains

$$\rho_A = \sum_{i,k,j} (A_\psi)_{ij} (A_\psi)^*_{kj} \, |i\rangle\langle k|. \tag{A.44}$$

Therefore, in matrix form,

$$\rho_A = A_\psi A_\psi^\dagger. \tag{A.45}$$

Similarly,

$$\rho_B = A_\psi^\dagger A_\psi. \tag{A.46}$$

Pure-State Concurrence

For pure two-qubit states, concurrence is defined as

$$C(|\psi\rangle) = |\langle\psi|(\sigma_y \otimes \sigma_y)|\psi^*\rangle| \,. \tag{A.47}$$

Using the computational-basis representation,

$$|\psi\rangle = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}, \qquad |\psi^*\rangle = \begin{pmatrix} a^* \\ b^* \\ c^* \\ d^* \end{pmatrix}, \tag{A.48}$$

one finds

$$(\sigma_y \otimes \sigma_y)|\psi^*\rangle = \begin{pmatrix} -d^* \\ c^* \\ b^* \\ -a^* \end{pmatrix}. \tag{A.49}$$

Taking the inner product yields

$$\langle\psi|(\sigma_y \otimes \sigma_y)|\psi^*\rangle = 2(ad - bc)^*. \tag{A.50}$$

Therefore,

$$C(|\psi\rangle) = 2|ad - bc|. \tag{A.51}$$

Using

$$\det(A_\psi) = ad - bc, \tag{A.52}$$

the concurrence becomes

$$C(|\psi\rangle) = 2|\det(A_\psi)|. \tag{A.53}$$

Relation with Reduced Density Operators

Using Eq. (A.45),

$$\rho_A = A_\psi A_\psi^\dagger, \tag{A.54}$$

one obtains

$$\det(\rho_A) = \det(A_\psi)\det(A_\psi^\dagger) = |\det(A_\psi)|^2. \tag{A.55}$$

Substituting into Eq. (A.53) gives

$$C(|\psi\rangle) = 2\sqrt{\det(\rho_A)}. \tag{A.56}$$

This reproduces the relation introduced in Chapter 4 [see Eq. (1.37)].

An identical relation holds for subsystem B:

$$C(|\psi\rangle) = 2\sqrt{\det(\rho_B)}. \tag{A.57}$$

These expressions show that bipartite entanglement is directly encoded in the reduced density operators.

Relation with Purity

For a single-qubit density operator ρ, one has

$$\mathrm{Tr}(\rho) = 1, \tag{A.58}$$

and purity

$$\gamma(\rho) = \mathrm{Tr}(\rho^2). \tag{A.59}$$

For 2×2 density operators, the determinant satisfies

$$\det(\rho) = \frac{1 - \gamma(\rho)}{2}. \tag{A.60}$$

Substituting Eq. (A.60) into Eq. (1.37) yields

$$C(|\psi\rangle) = \sqrt{2\,(1 - \gamma(\rho_A))}. \tag{A.61}$$

Therefore, for pure bipartite states, the mixedness of the reduced subsystem provides a direct quantitative measure of entanglement.

Separable states produce pure reduced density operators and therefore zero concurrence, whereas maximally entangled states produce maximally mixed reduced states and maximal concurrence.

B | Basis Representation and Invariance Properties

B.1. Concurrence: Basis Dependence, Representation Dependence, and Invariance

This appendix clarifies the distinction between representation-dependent quantities and physically invariant quantities in quantum mechanics, with particular emphasis on the concurrence used throughout the present work.

A potential conceptual ambiguity arises from the standard definition of concurrence for mixed two-qubit states, since the construction explicitly involves the operator σ_y and complex conjugation in the computational basis. At first sight, this may suggest that concurrence depends on a preferred polarization axis or on a specific basis representation.

The purpose of this appendix is to show that this interpretation is incorrect, and that concurrence constitutes an intrinsic, basis-independent measure of bipartite entanglement.

Representation Dependence and Basis Transformations

In quantum mechanics, vectors and operators are represented through coordinates and matrices relative to a chosen basis.

For a single qubit, a generic pure state may be written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

The coefficients α and β depend on the chosen basis representation.

For example, switching from the computational basis

$$\{|0\rangle, |1\rangle\}$$

to the rotated basis

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}},$$

changes the numerical coordinates of the state vector, while leaving the physical quantum state unchanged.

Similarly, a density operator transforms under a basis change according to

$$\rho \to \rho' = U\rho U^\dagger, \tag{B.1}$$

where U is a unitary change-of-basis operator.

The individual matrix entries of ρ therefore depend on the chosen representation.

However, physically meaningful invariant quantities such as

$$\mathrm{Tr}(\rho), \qquad \mathrm{Tr}(\rho^2), \qquad \det(\rho), \qquad \mathrm{spec}(\rho)$$

remain unchanged under Eq. (B.1).

This distinction is fundamental:

- matrix elements and coordinate representations are generally basis dependent,
- intrinsic physical properties are basis independent.

Direction Dependence and Physical Observables

A separate concept is the dependence of observables on physical measurement directions.

Expectation values such as

$$\langle\sigma_x\rangle, \qquad \langle\sigma_y\rangle, \qquad \langle\sigma_z\rangle$$

correspond to measurements performed along different axes of the Bloch sphere.

These quantities are physically direction dependent, since changing the measurement axis changes the measured observable.

For instance,

$$\langle \sigma_z \rangle = \mathrm{Tr}(\rho \sigma_z)$$

measures polarization along the computational Z-axis, while

$$\langle \sigma_x \rangle = \mathrm{Tr}(\rho \sigma_x)$$

measures polarization along the transverse X-axis.

Direction dependence therefore refers to the physical choice of measurement axis, rather than to the mathematical choice of representation basis.

These two notions are related but conceptually distinct.

Representation Dependence of Pauli Matrices

The Pauli operators themselves admit different matrix representations depending on the chosen basis.

In the computational basis,

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{B.2}$$

Under a basis transformation U, the matrix representation changes according to

$$\sigma_y \to \sigma_y' = U \sigma_y U^\dagger. \tag{B.3}$$

Consequently, the matrix entries appearing in Eq. (B.2) are representation dependent.

Importantly, this does not imply that the physical observable itself changes. Rather, only its coordinate representation changes.

This situation is directly analogous to ordinary vector coordinates in Euclidean geometry.

Concurrence and the Spin-Flip Construction

For mixed two-qubit states, concurrence is commonly expressed through the spin-flipped density operator

$$\tilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y), \tag{B.4}$$

where ρ^* denotes complex conjugation in the computational basis.

The concurrence is then defined as

$$C(\rho) = \max\left(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\right), \tag{B.5}$$

where λ_i are the ordered square roots of the eigenvalues of

$$\rho\tilde{\rho}.$$

At first sight, Eq. (B.4) may appear to privilege the Y-direction of the Bloch sphere because of the explicit presence of σ_y.

However, in this context, σ_y is not used as a measurement operator associated with the physical Y-axis. Instead, it appears because it realizes the invariant antisymmetric structure of two-dimensional spinor spaces.

Antisymmetric Structure of Two-Qubit States

Consider a generic pure two-qubit state

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle. \tag{B.6}$$

The coefficients may be organized into the matrix

$$\Psi = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \tag{B.7}$$

For pure states, concurrence reduces to

$$C(|\psi\rangle) = 2|ad - bc|. \tag{B.8}$$

The quantity

$$ad - bc$$

is precisely the determinant of the coefficient matrix Ψ:

$$\det(\Psi) = ad - bc. \tag{B.9}$$

This determinant originates from the antisymmetric bilinear form

$$\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tag{B.10}$$

which satisfies

$$\sigma_y = -i\varepsilon. \tag{B.11}$$

Thus, the appearance of σ_y in the concurrence construction is not associated with a preferred physical polarization axis, but with the antisymmetric invariant structure of two-level quantum systems.

Basis Independence of Concurrence

An entanglement measure must remain invariant under local unitary transformations, since local polarization rotations cannot create or destroy entanglement.

For any local unitary evolution,

$$\rho \to (U_A \otimes U_B)\rho(U_A^\dagger \otimes U_B^\dagger), \tag{B.12}$$

the concurrence satisfies

$$C\left[(U_A \otimes U_B)\rho(U_A^\dagger \otimes U_B^\dagger)\right] = C(\rho). \tag{B.13}$$

This invariance is essential for the physical interpretation of concurrence as an intrinsic measure of bipartite entanglement.

Consequently:

- the matrix representation of the spin-flipped operator depends on the chosen basis,
- the intermediate construction in Eq. (B.4) is representation dependent,
- the final concurrence $C(\rho)$ is basis independent.

Manifestly Basis-Independent Formulations

For pure states, concurrence admits formulations that make the invariance more transparent.

Using the reduced density operator

$$\rho_A = \mathrm{Tr}_{\mathrm{B}} \left(|\psi\rangle\langle\psi| \right),$$

one obtains

$$C(|\psi\rangle) = 2\sqrt{\det(\rho_A)}. \tag{B.14}$$

Equivalently,

$$C(|\psi\rangle) = \sqrt{2\left(1 - \mathrm{Tr}(\rho_A^2)\right)}. \tag{B.15}$$

These expressions involve only invariant quantities such as determinants and traces, and therefore do not rely explicitly on any preferred basis representation.

More fundamentally, concurrence for mixed states is defined through the convex-roof construction

$$C(\rho) = \min_{\{p_k, |\psi_k\rangle\}} \sum_k p_k C(|\psi_k\rangle), \tag{B.16}$$

where

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|. \tag{B.17}$$

This formulation is intrinsically basis independent.

The spin-flip formalism of Eq. (B.4) constitutes a special analytical construction that allows the explicit evaluation of the convex-roof optimization for two-qubit systems.

Physical Interpretation in Fiber-Based Quantum Channels

The distinction between representation-dependent quantities and invariant quantities plays a central role in the modeling strategy developed throughout the present work.

Local unitary transformations induced by fiber propagation,

$$U_A \otimes U_B,$$

may alter:

- local Bloch coordinates,
- Stokes parameters,
- tensor components,
- fixed-basis Bell-state fidelities,
- correlation functions evaluated along specific axes.

However, these transformations do not modify the intrinsic entanglement content of the bipartite state.

Consequently, the concurrence remains unchanged under coherent local polarization evolution.

By contrast, non-unitary effects such as ensemble averaging, decoherence, or polarization-mode dispersion generate effective mixed states and therefore reduce concurrence.

This distinction establishes the conceptual transition

$$\text{coherent local evolution} \rightarrow \text{effective decoherence} \rightarrow \text{entanglement degradation},$$

which constitutes one of the central physical mechanisms investigated in the simulator developed in this work.

C | Quantum Channels and Effective Fiber Models

C.1. Global Depolarizing Channel Analytics

This appendix derives the analytical expressions used to validate the depolarizing-channel experiment in Section **??**.

The reference state is the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + |10\rangle\right),$$

with density operator

$$\rho_0 = |\Psi^+\rangle\langle\Psi^+|.$$

The depolarized family is defined as

$$\rho(p) = (1-p)\rho_0 + \frac{p}{4} I_4, \qquad p \in [0,1]. \tag{C.1}$$

Density matrix representation

In the computational basis

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\},$$

the Bell-state density operator is

$$\rho_0 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Substituting this expression into Eq. (C.1) gives

$$\rho(p) = \begin{pmatrix} \frac{p}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{2} - \frac{p}{4} & \frac{1-p}{2} & 0 \\ 0 & \frac{1-p}{2} & \frac{1}{2} - \frac{p}{4} & 0 \\ 0 & 0 & 0 & \frac{p}{4} \end{pmatrix}.$$

Global purity

The global purity is defined as

$$\gamma_{AB}(p) = \mathrm{Tr}\left(\rho(p)^2\right).$$

Since ρ_0 is a rank-one projector and $I_4/4$ is proportional to the identity, the Bell-state direction remains an eigenvector of $\rho(p)$, with eigenvalue

$$\lambda_1 = 1 - \frac{3p}{4}.$$

The three orthogonal Bell-basis directions have eigenvalues

$$\lambda_2 = \lambda_3 = \lambda_4 = \frac{p}{4}.$$

Therefore,

$$\gamma_{AB}(p) = \left(1 - \frac{3p}{4}\right)^2 + 3\left(\frac{p}{4}\right)^2.$$

After simplification,

$$\gamma_{AB}(p) = 1 - \frac{3}{2}p + \frac{3}{4}p^2. \tag{C.2}$$

Reduced density operators

For the Bell state,

$$\mathrm{Tr}_B(\rho_0) = \mathrm{Tr}_A(\rho_0) = \frac{I_2}{2}.$$

Moreover,

$$\mathrm{Tr}_B\left(\frac{I_4}{4}\right) = \mathrm{Tr}_A\left(\frac{I_4}{4}\right) = \frac{I_2}{2}.$$

By linearity of the partial trace,

$$\rho_A(p) = \mathrm{Tr}_B(\rho(p)) = (1-p)\frac{I_2}{2} + p\frac{I_2}{2} = \frac{I_2}{2}.$$

Similarly,

$$\rho_B(p) = \frac{I_2}{2}.$$

The reduced purities are therefore

$$\gamma_A(p) = \gamma_B(p) = \mathrm{Tr}\left[\left(\frac{I_2}{2}\right)^2\right] = \frac{1}{2}. \tag{C.3}$$

Fidelity with respect to $|\Psi^+\rangle$

Since the reference state is pure, the fidelity is

$$F_{\Psi^+}(p) = \langle\Psi^+|\rho(p)|\Psi^+\rangle.$$

Substituting Eq. (C.1),

$$F_{\Psi^+}(p) = (1-p)\langle\Psi^+|\rho_0|\Psi^+\rangle + \frac{p}{4}\langle\Psi^+|I_4|\Psi^+\rangle.$$

Since

$$\langle\Psi^+|\rho_0|\Psi^+\rangle = 1, \qquad \langle\Psi^+|I_4|\Psi^+\rangle = 1,$$

one obtains

$$F_{\Psi^+}(p) = 1 - \frac{3}{4}p. \tag{C.4}$$

Concurrence

The density operator $\rho(p)$ is Bell-diagonal. For a Bell-diagonal two-qubit state, the concurrence is

$$C(p) = \max\left(0, 2\lambda_{\max} - 1\right),$$

where $\lambda_{\max}$ is the largest Bell-basis eigenvalue.

In the present case,

$$\lambda_{\max} = 1 - \frac{3p}{4}.$$

Therefore,

$$C(p) = \max\left(0, 1 - \frac{3p}{2}\right). \tag{C.5}$$

The concurrence vanishes when

$$1 - \frac{3p}{2} = 0,$$

which gives the separability threshold

$$p = \frac{2}{3}. \tag{C.6}$$

Correlation tensor

The Pauli correlations are defined as

$$c_{ij}(p) = \mathrm{Tr}\left[\rho(p)\left(\sigma_i \otimes \sigma_j\right)\right], \qquad i, j \in \{x, y, z\}.$$

Using linearity,

$$c_{ij}(p) = (1-p)\,\mathrm{Tr}\left[\rho_0\left(\sigma_i \otimes \sigma_j\right)\right] + \frac{p}{4}\,\mathrm{Tr}\left[I_4\left(\sigma_i \otimes \sigma_j\right)\right].$$

Since every Pauli product $\sigma_i \otimes \sigma_j$ with $i, j \in \{x, y, z\}$ is traceless,

$$\mathrm{Tr}\left[I_4\left(\sigma_i \otimes \sigma_j\right)\right] = 0.$$

Thus,

$$c_{ij}(p) = (1-p)c_{ij}(0).$$

Equivalently, the full correlation tensor is uniformly contracted:

$$T(p) = (1-p)T_{\Psi^+}. \tag{C.7}$$

For the reference Bell state $|\Psi^+\rangle$,

$$T_{\Psi^+} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Therefore,

$$T(p) = \begin{pmatrix} 1-p & 0 & 0 \\ 0 & 1-p & 0 \\ 0 & 0 & -(1-p) \end{pmatrix}. \tag{C.8}$$

The diagonal correlations are

$$c_{xx}(p) = 1-p, \qquad c_{yy}(p) = 1-p, \qquad c_{zz}(p) = -(1-p). \tag{C.9}$$

Summary

Collecting the analytical expressions:

$$\boxed{\begin{aligned}
\gamma_{AB}(p) &= 1 - \frac{3}{2}p + \frac{3}{4}p^2, \\
\gamma_A(p) &= \gamma_B(p) = \frac{1}{2}, \\
F_{\Psi^+}(p) &= 1 - \frac{3}{4}p, \\
C(p) &= \max\left(0, 1 - \frac{3}{2}p\right), \\
c_{xx}(p) &= 1 - p, \qquad c_{yy}(p) = 1 - p, \qquad c_{zz}(p) = -(1 - p).
\end{aligned}}$$

These expressions provide the analytical reference used for the numerical validation of the depolarizing-channel experiment.

C.2. Commutation of Local Phase Evolution and Isotropic Local Depolarization

This appendix establishes explicitly the commutation relation underlying the composite phase–depolarization fiber model introduced in Section 5.4.

In particular, it is shown that the reduced phase-evolution model considered in this work commutes with isotropic local depolarizing channels acting independently on the two subsystems. This property justifies the equivalence between applying local depolarization before or after coherent phase evolution within the isotropic approximation adopted in the present work.

Effective Channel Structure

The coherent phase evolution acting on subsystem A is represented through the unitary operator

$$U_\theta = E_\theta \otimes I, \tag{C.10}$$

where

$$E_\theta = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}. \tag{C.11}$$

The corresponding unitary channel is

$$\mathcal{U}_\theta(\rho) = U_\theta \rho U_\theta^\dagger. \tag{C.12}$$

Local isotropic depolarization acting independently on the two subsystems is described through

$$\mathcal{E}_{p_A,p_B} = \mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}, \tag{C.13}$$

where the single-qubit depolarizing channel satisfies

$$\mathcal{D}_p(I) = I, \tag{C.14}$$

and

$$\mathcal{D}_p(\sigma_i) = (1-p)\sigma_i, \qquad i \in \{x, y, z\}. \tag{C.15}$$

The objective is to prove that

$$\mathcal{E}_{p_A,p_B} \circ \mathcal{U}_\theta = \mathcal{U}_\theta \circ \mathcal{E}_{p_A,p_B}. \tag{C.16}$$

Pauli-Basis Expansion

Consider a generic two-qubit density operator expanded in the Pauli basis:

$$\rho = \frac{1}{4} \sum_{\mu,\nu=0}^{3} S_{\mu\nu}\, \sigma_\mu \otimes \sigma_\nu, \tag{C.17}$$

where

$$\sigma_0 = I,$$

and

$$\sigma_1 = \sigma_x, \qquad \sigma_2 = \sigma_y, \qquad \sigma_3 = \sigma_z.$$

Since both channels are linear, it is sufficient to study their action on generic Pauli-basis elements

$$\sigma_\mu \otimes \sigma_\nu.$$

Action of the Unitary Channel

The local phase unitary acts only on subsystem A. Therefore,

$$\mathcal{U}_\theta \left(\sigma_\mu \otimes \sigma_\nu\right) = \left(E_\theta \sigma_\mu E_\theta^\dagger\right) \otimes \sigma_\nu. \tag{C.18}$$

For $\mu = 0$, the identity operator is preserved trivially:

$$E_\theta I E_\theta^\dagger = I. \tag{C.19}$$

For $\mu \in \{1, 2, 3\}$, the transformed operator admits a Pauli-basis expansion of the form

$$E_\theta \sigma_\mu E_\theta^\dagger = a\sigma_0 + b\sigma_1 + c\sigma_2 + d\sigma_3. \tag{C.20}$$

However, unitary conjugation preserves tracelessness:

$$\operatorname{Tr}\left(E_\theta \sigma_\mu E_\theta^\dagger\right) = \operatorname{Tr}(\sigma_\mu) = 0, \qquad \mu \in \{1, 2, 3\}. \tag{C.21}$$

Since

$$\operatorname{Tr}(\sigma_0) = 2, \qquad \operatorname{Tr}(\sigma_i) = 0 \quad (i = 1, 2, 3),$$

Eq. (C.20) implies

$$a = 0.$$

Therefore,

$$E_\theta \sigma_\mu E_\theta^\dagger = b\sigma_1 + c\sigma_2 + d\sigma_3, \qquad \mu \in \{1,2,3\}. \tag{C.22}$$

The unitary evolution therefore preserves the traceless Pauli subspace.

Comparison of the Two Channel Orders

Consider first the ordering

$$\mathcal{E}_{p_A,p_B} \circ \mathcal{U}_\theta. \tag{C.23}$$

Acting on a generic basis element gives

$$\begin{aligned}&\mathcal{E}_{p_A,p_B}\left[\mathcal{U}_\theta\left(\sigma_\mu \otimes \sigma_\nu\right)\right]\\ &= \mathcal{D}_{p_A}\left(E_\theta \sigma_\mu E_\theta^\dagger\right) \otimes \mathcal{D}_{p_B}\left(\sigma_\nu\right).\end{aligned} \tag{C.24}$$

Using Eq. (C.22), the first factor contains only traceless Pauli operators. Consequently,

$$\begin{aligned}&\mathcal{D}_{p_A}\left(E_\theta \sigma_\mu E_\theta^\dagger\right)\\ &= (1-p_A)\left(E_\theta \sigma_\mu E_\theta^\dagger\right), \qquad \mu \in \{1,2,3\}.\end{aligned} \tag{C.25}$$

Now consider the opposite ordering:

$$\mathcal{U}_\theta \circ \mathcal{E}_{p_A,p_B}. \tag{C.26}$$

Its action gives

$$\begin{aligned}&\mathcal{U}_\theta\left[\mathcal{E}_{p_A,p_B}\left(\sigma_\mu \otimes \sigma_\nu\right)\right]\\ &= E_\theta \mathcal{D}_{p_A}(\sigma_\mu) E_\theta^\dagger \otimes \mathcal{D}_{p_B}(\sigma_\nu).\end{aligned} \tag{C.27}$$

Using Eq. (C.15),

$$E_\theta \mathcal{D}_{p_A}(\sigma_\mu) E_\theta^\dagger = (1 - p_A) E_\theta \sigma_\mu E_\theta^\dagger. \tag{C.28}$$

Comparing Eqs. (C.25) and (C.28) shows that the two orderings coincide.

Therefore,

$$\mathcal{E}_{p_A,p_B} \circ \mathcal{U}_\theta = \mathcal{U}_\theta \circ \mathcal{E}_{p_A,p_B}. \tag{C.29}$$

Physical Interpretation

The commutation property derived above follows from two structural features of the effective model:

- the isotropic depolarizing channel acts uniformly on the traceless Pauli subspace;
- unitary conjugation preserves the traceless structure of Pauli operators.

Consequently, coherent phase evolution rotates the polarization-correlation structure without modifying its magnitude, while isotropic local depolarization contracts the correlation amplitudes uniformly without introducing preferred directions in polarization space.

Within the isotropic approximation adopted in the present work, the order of coherent phase evolution and local depolarization is therefore irrelevant.

This property considerably simplifies the analytical structure of the composite phase–depolarization fiber model introduced in Section 5.4, since coherent phase evolution and isotropic local depolarization may be treated independently while remaining mathematically equivalent under composition.

C.3. Statistical Composite Fiber-Channel Analytics

This appendix derives the analytical expressions associated with the statistical composite fiber model introduced in Section **??**.

The model combines:

- statistical phase averaging,
- local isotropic depolarization acting independently on the two propagation arms.

The reference Bell state is

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right),$$

with density operator

$$\rho_0 = |\Psi^+\rangle\langle\Psi^+|.$$

The phase-evolved pure state is

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + e^{i\theta}|10\rangle\right). \tag{C.30}$$

Define the phase-coherence parameter

$$\mu = \left\langle e^{i\theta}\right\rangle = \sum_k p_k e^{i\theta_k}, \tag{C.31}$$

and the depolarization contraction factor

$$\eta = (1-p_A)(1-p_B). \tag{C.32}$$

Ensemble density operator

The ensemble-averaged density operator is

$$\rho^{(\mathrm{ens})} = \sum_k p_k |\psi(\theta_k)\rangle\langle\psi(\theta_k)|. \tag{C.33}$$

Using Eq. (C.30), one obtains

$$\rho^{(\mathrm{ens})} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & \mu^* & 0 \\ 0 & \mu & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{C.34}$$

Including local isotropic depolarization contracts the bipartite correlations by the factor

η. The resulting effective state is

$$\rho_\eta^{(\mathrm{ens})} = \frac{1}{4}\left[I \otimes I + \sum_{i,j} T_{ij}^{(\mathrm{ens})} \sigma_i \otimes \sigma_j\right], \tag{C.35}$$

where the tensor components are derived below.

Correlation tensor

For the ensemble state without local depolarization, the correlation tensor is

$$T^{(\mathrm{ens})} = \begin{pmatrix} \mathrm{Re}(\mu) & -\mathrm{Im}(\mu) & 0 \\ \mathrm{Im}(\mu) & \mathrm{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{C.36}$$

Under local isotropic depolarization,

$$T_\eta^{(\mathrm{ens})} = \eta T^{(\mathrm{ens})}, \tag{C.37}$$

thus

$$T_\eta^{(\mathrm{ens})} = \eta \begin{pmatrix} \mathrm{Re}(\mu) & -\mathrm{Im}(\mu) & 0 \\ \mathrm{Im}(\mu) & \mathrm{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{C.38}$$

The tensor components are therefore

$$T_{xx} = \eta\,\mathrm{Re}(\mu), \tag{C.39}$$
$$T_{xy} = -\eta\,\mathrm{Im}(\mu), \tag{C.40}$$
$$T_{yx} = \eta\,\mathrm{Im}(\mu), \tag{C.41}$$
$$T_{yy} = \eta\,\mathrm{Re}(\mu), \tag{C.42}$$
$$T_{zz} = -\eta. \tag{C.43}$$

All remaining tensor components vanish.

Global purity

The global purity is

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \mathrm{Tr}\left[\left(\rho_{AB}^{(\mathrm{ens})}\right)^2\right]. \tag{C.44}$$

Using the Pauli-basis representation,

$$\rho = \frac{1}{4}\left[I \otimes I + \sum_{i,j} T_{ij}\sigma_i \otimes \sigma_j\right], \tag{C.45}$$

together with Pauli orthogonality,

$$\mathrm{Tr}\left[(\sigma_i \otimes \sigma_j)(\sigma_k \otimes \sigma_l)\right] = 4\delta_{ik}\delta_{jl}, \tag{C.46}$$

one obtains

$$\gamma = \frac{1}{4}\left[1 + \sum_{i,j} T_{ij}^2\right]. \tag{C.47}$$

Substituting Eq. (C.38),

$$\sum_{i,j} T_{ij}^2 = \eta^2\left[2|\mu|^2 + 1\right]. \tag{C.48}$$

Therefore,

$$\boxed{\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1 + \eta^2(1 + 2|\mu|^2)}{4}} \tag{C.49}$$

For $\eta = 1$, one recovers

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1 + |\mu|^2}{2}. \tag{C.50}$$

Reduced density operators

The reduced density operators remain maximally mixed:

$$\rho_A = \rho_B = \frac{I_2}{2}. \tag{C.51}$$

Therefore,

$$\boxed{\gamma_A = \gamma_B = \frac{1}{2}}. \tag{C.52}$$

Fidelity with respect to $|\Psi^+\rangle$

The Bell-state fidelity is

$$F_{\Psi^+} = \langle \Psi^+ | \rho_\eta^{(\mathrm{ens})} | \Psi^+ \rangle. \tag{C.53}$$

Using the depolarized ensemble density operator,

$$F_{\Psi^+} = \frac{1-\eta}{4} + \eta \frac{1 + \mathrm{Re}(\mu)}{2}. \tag{C.54}$$

Therefore,

$$\boxed{F_{\Psi^+} = \frac{1-\eta}{4} + \eta \frac{1 + \mathrm{Re}(\mu)}{2}} \tag{C.55}$$

Equivalently,

$$F_{\Psi^+} = \frac{1+\eta}{4} + \frac{\eta}{2} \mathrm{Re}(\mu). \tag{C.56}$$

For $\eta = 1$, this reduces to

$$F_{\Psi^+} = \frac{1 + \mathrm{Re}(\mu)}{2}. \tag{C.57}$$

For $\eta = 0$, the state is fully mixed and one obtains

$$F_{\Psi^+} = \frac{1}{4}. \tag{C.58}$$

Concurrence

The ensemble density operator belongs to the class of X-states. For a two-qubit X-state,

$$C = 2\max\left(0, |\rho_{23}| - \sqrt{\rho_{11}\rho_{44}}\right). \tag{C.59}$$

For the present model,

$$\rho_{23} = \frac{\eta\mu}{2}, \tag{C.60}$$

while the depolarizing contribution generates

$$\rho_{11} = \rho_{44} = \frac{1-\eta}{4}. \tag{C.61}$$

Therefore,

$$\begin{aligned} C &= 2\max\left[0, \frac{\eta|\mu|}{2} - \frac{1-\eta}{4}\right] \\ &= \max\left[0, \eta|\mu| - \frac{1-\eta}{2}\right]. \end{aligned} \tag{C.62}$$

Hence,

$$\boxed{C = \max\left[0, \eta|\mu| - \frac{1-\eta}{2}\right]} \tag{C.63}$$

CHSH nonlocality

The maximal CHSH parameter is determined by the Horodecki criterion:

$$S_{\max} = 2\sqrt{\lambda_1 + \lambda_2}, \tag{C.64}$$

where λ_1, λ_2 are the two largest eigenvalues of

$$T^T T.$$

Using Eq. (C.38),

$$T^T T = \eta^2 \begin{pmatrix} |\mu|^2 & 0 & 0 \\ 0 & |\mu|^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{C.65}$$

The two largest eigenvalues are therefore

$$\lambda_1 = \eta^2, \qquad \lambda_2 = \eta^2 |\mu|^2.$$

Thus,

$$\boxed{S_{\max} = 2\eta\sqrt{1 + |\mu|^2}} \tag{C.66}$$

Bell-inequality violation requires

$$S_{\max} > 2, \tag{C.67}$$

which gives

$$\boxed{\eta\sqrt{1 + |\mu|^2} > 1} \tag{C.68}$$

Summary

Collecting the analytical expressions:

$$\boxed{\begin{aligned} T_{xx} &= \eta\,\mathrm{Re}(\mu), \\ T_{xy} &= -\eta\,\mathrm{Im}(\mu), \\ T_{yx} &= \eta\,\mathrm{Im}(\mu), \\ T_{yy} &= \eta\,\mathrm{Re}(\mu), \\ T_{zz} &= -\eta, \\ \gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) &= \frac{1+\eta^2(1+2|\mu|^2)}{4}, \\ \gamma_A = \gamma_B &= \frac{1}{2}, \\ F_{\Psi^+} &= \frac{1-\eta}{4} + \eta\frac{1+\mathrm{Re}(\mu)}{2}, \\ C &= \max\left[0, \eta|\mu| - \frac{1-\eta}{2}\right], \\ S_{\max} &= 2\eta\sqrt{1+|\mu|^2}. \end{aligned}}$$

These expressions provide the analytical reference for the statistical composite fiber-channel model introduced in Section **??**.

List of Figures

List of Tables